Do Distributional Concerns Justify Lower Environmental Taxes?

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Abstract

Should externality taxes be reduced if they are regressive? We introduce a model in which the government raises revenue from distortionary income and commodity taxes. Whether the tax on the externality should be adjusted away from the Pigouvian level of marginal damage depends on why households with lower incomes are more carbon-intensive in their consumption. Differences in demand due to preference heterogeneity across the income distribution change the optimal tax, while differences due to income effects do not. We derive sufficient statistics for optimal policy, and use them to study carbon taxation in the United States. Our empirical results suggest an optimal carbon tax that is remarkably close to the Pigouvian level, but with higher carbon taxes for very high-income households if feasible. When we allow for heterogeneity in preferences at each income level as well as across the income distribution, our optimal tax schedules move closer to the Pigouvian benchmark.

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1 Introduction

Human-generated carbon emissions are widely accepted to be causing large social costs, both now and in the future (Intergovernmental Panel on Climate Change, 2023). Economists view this as a classic case for government intervention, because there is an externality: Consumers and producers do not bear the full cost of the emissions they generate.

Most economists also agree that the solution is to impose a "Pigouvian" tax on goods that generate negative externalities—in this case, goods whose production or consumption generates carbon emissions (Akerlof et al., 2019).[1] Setting this tax equal to the marginal damage to others from emissions ensures that consumers and producers pay the true social costs of their choices (Pigou, 1920; Sandmo, 1975). This Pigouvian tax would be optimal if non-distortionary lump sum compensation of winners and losers were possible. However, when redistribution is costly, it is much less clear how a carbon tax should be set given the potential tradeoff between externality correction and distributional impacts.

The fact that carbon taxes are regressive may be an important reason why they are politically unpopular (Hassett et al., 2009; Grainger and Kolstad, 2010; Mathur and Morris, 2014; Metcalf, 2009; Williams III et al., 2015; Pizer and Sexton, 2019). There is widespread concern about their distributional impacts (Dietz and Atkinson, 2010; Dechezleprêtre et al., 2025). As we confirm in our data, this is justified. A uniform tax on all carbon-generating activities would lead poorer households to pay more as a share of their incomes (or expenditures) than richer households. This raises the question of whether corrective taxes should be lowered relative to the Pigouvian benchmark for distributional reasons.

We answer that question. Specifically, we ask how the optimal carbon tax changes if society values equality but compensation can only occur using efficiency-reducing income and commodity taxes. The distributional impact of a carbon tax cannot readily be dismissed in this setting. Imposing the tax produces winners and losers. Compensating the losers by taxing the winners could lead to a greater loss in efficiency than the gain from correcting the externality in the first place. In other cases, the combined package of carbon taxation and compensation could lead to net efficiency gains. Only in the knife-edge case where any two households would allocate their spending in precisely the same way if they had the same income is the carbon tax optimally set at the Pigouvian level of marginal damage (Kaplow, 2012). This is the case in which the "first best" remains optimal even in a "second best" world. By contrast, if households differ in how they would choose to spend a given level of income, the optimal tax changes because the planner faces a trade-off between the efficiency of redistribution and the extent of externality correction.

[1] Panelists in the Initiative on Global Markets (IGM) Economic Experts Panel have consistently endorsed carbon taxation as a preferred climate policy (IGM Forum, 2011, 2012, 2018, 2025).

Our contribution is to provide a practical and general characterization of how a corrective tax should be adjusted in light of its distributional impact. We start with a model in which each household chooses its income and carbon intensity. Building on work by Ferey et al. (2024), we use our model to derive sufficient statistics formulas that characterize the optimal trade-off between externality correction and redistribution. Heterogeneity in preferences for carbon-intensive consumption plays a pivotal role, both within and across income levels. Taking into account both types of heterogeneity, we apply our framework empirically to study the optimal taxation of electricity, natural gas, heating fuels, and gasoline consumption in the United States. Our results suggest that the optimal carbon tax on these goods remains close to the Pigouvian prescription.

We note that our focus on whether the carbon tax should be changed for distributional reasons is distinct from questions of how much to compensate households, which others discuss in detail (Grainger and Kolstad, 2010; Goulder et al., 2019). Our framework does incorporate flexible compensation, but the pattern of that compensation has no direct connection to the question of whether the carbon tax should differ from the Pigouvian level. Rather the optimal level of the carbon tax hinges on whether it is more efficient to reduce inequality by lowering the carbon tax versus increasing the income tax.

Preference heterogeneity is important because it determines the relative cost of redistribution via income versus carbon taxes. A carbon tax set at the Pigouvian level ensures that households spend their after-tax incomes efficiently. Starting from this level, raising the carbon tax distorts consumption choices because the externality is no longer exactly internalized. However, the reform also changes labor supply choices. Raising the carbon tax lowers labor supply, while the compensatory income tax reduction increases labor supply. If all households have identical preferences for carbon intensity, the two labor supply effects cancel out. This is because identical preferences imply that the combined reform to the income and carbon taxes exactly compensates every household for the impact of the carbon tax not only at their present income, but also at *every other* level of income they might consider. There is therefore no net impact on individuals' labor supply decisions, and no reason to deviate from purely Pigouvian taxation.

Heterogeneity in preferences breaks the neutrality of the combined reform. This is easiest to see in the simpler case in which households with different income levels vary in their preferences, but all households with the same income are equally carbon-intensive. If higher-income households have a relative preference for carbon intensity, so that they would be more carbon-intensive at any given level of disposable income, raising the carbon tax and lowering the income tax yields a net increase in labor supply. In this case, raising the carbon tax redistributes more efficiently than income taxation at the margin. The optimal tax on carbon is therefore greater than the marginal damage. By similar logic, if lower productivity types have more carbon-intensive preferences, lowering the carbon tax below the Pigouvian level becomes an efficient way to redistribute.

Our next step is to extend the model to allow for multidimensional heterogeneity in the sense that carbon intensity is allowed to vary across households with the same level of income. In the multidimensional case, adjustments to the income tax can no longer compensate every household for the introduction (or increase) of a carbon tax. The reason is simple: At a given income, households differ in how much of the carbon-intensive good they consume, so some require more compensation than others, and a single income-based adjustment cannot make all of them whole. We nonetheless derive an empirically tractable optimal tax condition. While the lessons from the unidimensional case carry over qualitatively, an important result is that variation in the causal effect of higher income on consumption choices tends to dampen any deviation of the optimal tax from its Pigouvian level. This force reflects imperfect targeting: because the planner cannot charge different carbon-tax rates to different types who earn the same income, the rate that is most efficient for one group leaves an inefficiently high (or low) rate for another.

Assuming that differences in preferences do justify deviating from the Pigouvian level of taxation, distributional concerns lead the government to trade off externality correction and redistribution. Our model highlights that the optimal tax depends on how responsive households are to the carbon tax and income tax. If the income tax is highly distortionary, it becomes more attractive to use the carbon tax to achieve distributional aims. Using the carbon tax to redistribute is also appealing if carbon intensity is inelastic to changes in the tax on carbon. Conversely, if changing the carbon tax sharply changes carbon emissions, using the carbon tax for redistribution is less desirable.

Unlike the classic Pigouvian case without distributional concerns, the optimal externality correction is not necessarily the same at all levels of consumption. If nonlinear taxation is feasible, for example through means-tested vouchers or nonlinear electricity and gas pricing, the optimal carbon tax may be progressive or regressive. There are many real-life examples of nonlinear pricing and taxation schemes such as electricity tiered pricing, congestion taxes, and emission-based vehicle registration fees or subsidies (see e.g., Zhang and Qin, 2015; ACEA, 2020). Whether such nonlinearity is optimal depends on the rate at which the preference for carbon intensity changes with income in different parts of the distribution. It does not depend on whether the carbon tax is regressive or progressive in the sense that poorer or richer households pay more as a share of their incomes.

Our final step is to apply our framework to carbon taxation in the United States. Our model highlights the need to empirically isolate differences in preferences between people with different levels of income. This is distinct from income effects, which can also drive differences in carbon intensity along the income distribution.[2] Indeed, it is clearly possible for any given cross-sectional relationship between income and carbon intensity to be explained exclusively by income effects, by preference heterogeneity, or by some combination of the two. The fact that poorer households are more carbon-intensive, and carbon taxes correspondingly regressive, does not therefore suffice to tell us whether the carbon tax should deviate from the Pigouvian benchmark.

We again start with the case where preferences vary across but not within income levels. Our model provides a convenient way to identify differences in preferences between households with different status quo incomes. Following Ferey et al. (2024), we compare two statistics. First, we measure the cross-sectional relationship between income and consumption of carbon-intensive goods. This captures differnces in consumption due to both income effects and preference heterogeneity. Second, we measure income effects by quantifying the causal impact of higher income on a given household's consumption of carbon-intensive goods. Finally, we subtract income effects from the cross-sectional relationship, leaving only the part that stems from preferences.

We implement this framework for the United States using survey evidence from 47,054 households in the Consumer Expenditure Surveys (CEX) and 1,448 respondents from a custom online survey. Our analysis focuses on the four expenditure categories that account for nearly all of spending on goods with *direct* household emissions, aggregated into a single "dirty good": gasoline, natural gas, electricity, and heating fuels.[3] In our survey, we ask respondents for information about their income, overall expenditure, and how they split that expenditure over these and other categories. We then ask how their expenditure on each of these categories would change if they had more to spend.

Our first step is to measure the cross-sectional relationship between household taxable income and consumption of carbon-intensive goods. While higher-income households consume a greater (absolute) quantity of these goods, they allocate a smaller share of their total consumption to carbon-intensive goods, reflecting an increasing but concave relationship between income and carbon-intensive good consumption.

Next, we measure respondents' causal consumption response to a small and permanent increase in household income across the income distribution. Surprisingly, the results

[2] The relevant distinction for our analysis is whether differences in choices by people with different incomes simply reflect differences in the amount they have to spend (income effects), or factors that are independent of income, which would drive different choices even conditional on income. The latter category includes true "preferences", but also habits or constraints that arise due to factors such as where one lives.

[3] Together, these four categories account for 94.5% of expenditures on goods with direct household carbon emissions, and for 97.4% of the associated emissions; the remainder corresponds to air travel.

indicate that the pattern of income effects in carbon intensity closely matches the cross-sectional variation in consumption in the Consumer Expenditure Survey. We note that this is not at all mechanical or obvious. Indeed, these causal consumption responses are within-individual effects, which are compared to variation across individuals. The result suggests that cross-sectional variation in consumption of these carbon-intensive goods is largely explained by income effects, and not by differences in preferences.

The interpretation of these survey responses hinges on the reliability of stated preferences. Growing evidence suggests that stated preferences perform well (List et al., 2006; Fuster and Zafar, 2023), yet reliability inevitably depends on context. In our setting, we have a way of verifying the reliability of our survey. Because we measure status quo allocations across different types of goods in both our survey and the Consumer Expenditure Survey, we can verify that the cross-sectional relationship between income and carbon-intensive good consumption is very similar across both sources. This lends credibility to our survey instrument and sample. We also note that we set the hypothetical income increase to be familiar and relevant for respondents (Fuster and Zafar, 2023).

Our empirical results allow us to estimate the optimal tax on carbon-intensive goods. Differences in preferences justify only small deviations from the Pigouvian prescription of setting the tax equal to marginal damage. Our estimates suggest that the Pigouvian level of the tax is about 40 cents per dollar of expenditure (assuming a Social Cost of Carbon of $200 per ton). Taking into account differences in preferences, we find that the optimal linear tax would be 1.2 cents above this. The very small magnitude of this adjustment is robust to a wide range of parameterizations, reflecting our core empirical result that the causal and cross-sectional relationships between income and carbon intensity match closely on average. Finally, we account for heterogeneity within as well as across income levels, which further compresses the optimal tax toward its Pigouvian level.

If taxes on externality-generating goods can be nonlinear—which has precedents—there are larger deviations from the Pigouvian prescription in some cases. For a nonlinear tax, we compare the cross-sectional and causal relationships between income and consumption at each income level. Consider households with incomes around $40,000. We find that the estimated causal consumption response is slightly flatter than the slope of the cross-sectional relationship between income and carbon intensity. This implies a positive, but small, preference-heterogeneity component, and justifies a carbon tax around 1.1 cents per dollar of expenditure above the Pigouvian level. Around $100,000, the estimated taste-heterogeneity component is slightly negative, so the optimal tax falls just below the Pigouvian level. Households with incomes closer to $250,000 would face carbon taxes of 45.7 cents, which is 5.7 cents (14.1%) higher than the Pigouvian level.

Our paper provides an empirical and theoretical framework that could be applied more broadly to environmental taxes, or any other tax designed to correct an externality. Consistent with the principle of additivity (Sandmo, 1975; Kopczuk, 2003), the total tax on any externality-generating activity is a combination of a "Pigouvian" correction and a distributional component. We provide a way to measure the distributional component and show how it depends on causal and cross-sectional sufficient statistics. The optimal tax on the externality may deviate positively or negatively from the Pigouvian prescription. In some cases, the total tax could even flip sign relative to the externality. In our case, we find that the efficiency gains from using the carbon tax to redistribute are small, so the optimal carbon tax stays very close to the Pigouvian level. However, applying our framework to other taxes may yield different results.

Connections in the Literature

Externalities have long been recognized as providing a motivation for government intervention. Corrective taxation as a solution dates back to Pigou (1920). Economists have studied wide varieties of externalities in practice, including environmental taxes (Weitzman, 1974; Andersson, 2019), occupation choice (Lockwood et al., 2017), congestion (Walters, 1961), labor market distortions (Craig, 2023), control of a pandemic (Craig and Hines, 2020), rent-seeking (Rothschild and Scheuer, 2016), and positional externalities (Aronsson and Johansson-Stenman, 2008).

A subset of this literature incorporates distributional concerns. Kopczuk (2003) shows how the optimal tax on an externality can be written as the sum of an externality correction component and a separate term capturing other objectives, building on Sandmo (1975). Kaplow (2012) studies a special case without preference heterogeneity, showing that the optimal tax is equal to marginal damage in this case. Pirttilä and Tuomala (1997) allow for interactions with labor supply. Jacobs and van der Ploeg (2019) focus on linear taxation. Cremer et al. (1998) and Cremer et al. (2003) consider optimal taxation and externality correction with a small number of types, demonstrating the importance of homogeneity and separability in preferences, highlighting the potential for the optimal tax on the externality-generating good to have the opposite sign from the externality. Ahlvik et al. (2024) focus on different types of equity considerations and differences in cost burdens. They use a mechanism design approach to study the interaction of income taxation and externality correction, including an empirical exercise with much stronger restrictions on individual heterogeneity than ours. Jacobs and de Mooij (2015) ask whether the Pigouvian correction should be adjusted for the marginal cost of public funds. Belfiori et al. (2024) focus on compensation in a macroeconomic model without heterogeneity in preferences for carbon intensity. Micheletto (2008) studies the correction of externalities that are not "atmospheric" so that externality correction is imperfectly targeted.

We build on an approach to optimal commodity taxation introduced by Allcott et al. (2019), which Ferey et al. (2024) later generalized and illustrated using savings taxation.[4] The authors highlight the core insight that causal income effects can be subtracted from cross-sectional variation in consumption to measure the degree to which preferences vary across income levels. We apply these ideas to shed light on how externality-correcting taxes should change with preference heterogeneity, including multidimensional heterogeneity; we then apply it to study carbon emissions by households, which is an important example of an externality. These papers and ours build on previous work (Saez, 2002; Piketty and Saez, 2013; Diamond and Spinnewijn, 2011; Gauthier and Henriet, 2018); and connect to concurrent theoretical analysis by Doligalski et al. (2025).

Empirically, recent studies highlight significant heterogeneity in consumer preferences for carbon-intensive goods vs. cleaner alternatives. For example, Lyubich (2024) decomposes the unobserved heterogeneity in US household carbon consumption and finds, by and large, a greater role for person effects (up to 50% of the variance decomposition) relative to place effects (up to 23%). Similarly, Dorsey and Wolfson (2024) provide evidence of across-income preference heterogeneity in solar panel uptake.

Our focus is on heterogeneity between households and the importance of the distortionary costs of redistributive taxation. We do not incorporate general equilibrium effects or dynamics, which can provide important additional motives for deviating from standard Pigouvian taxation. This has been considered in other work (Cremer and Gahvari, 2001; Cremer et al., 2010; Douenne et al., 2025; Barrage, 2020; Bierbrauer, 2024; Fried et al., 2024). Similarly, we do not consider multinational issues (see Aronsson and Blomquist, 2003) or explicitly model multiple generations (Fried et al., 2018). Our work complements Fricke et al. (2025), who study the optimal rebate when carbon emissions are capped.

2 Externality Correction With Taste Heterogeneity

Consider a population of individuals indexed by type $w \in W = \mathbb{R}_{++}$, where w has differentiable cumulative distribution F. Individual w chooses taxable income $z \in \mathbb{R}_{+}$. Income has differentiable cumulative distribution $H_z(z)$ and density $h_z(z)$. Income is subject to nonlinear income taxation with tax schedule $T_z(z)$. She allocates her after-tax income, $z - T_z(z)$, between a numeraire consumption good, c, and an externality-generating good, x. A tax, $T_x(x)$, must be paid on consumption of the externality-generating good. We

[4]Both of these papers also discuss corrective taxation. In Allcott et al. (2019), the corrective taxes primarily correct internalities; in Ferey et al. (2024), corrective motives—arising from behavioral biases or externalities—are introduced as an extension, though the paper primarily focuses on savings taxation.

constrain $T_z(z)$ and $T_x(x)$ to be twice continuously differentiable.

We focus in this section on the case in which heterogeneity is unidimensional in the sense that $w \in W \subset \mathbb{R}$ and utility is $u(c, x, z \mid w)$. This implies that there is no variation in preferences conditional on w. However, individuals with different w may have different preferences over the consumption goods in the sense that they would prefer different consumption bundles even if they were forced to have the same z and $z - T(z)$. In Section 3, we further extend our results to allow for multidimensional heterogeneity.

Separable Benchmark. As an important benchmark, we will consider the case, as in Atkinson and Stiglitz (1976), where preferences over consumption goods are weakly separable from labor supply, and homogeneous across agents, including those of different types. In this special case, it is well-known that we obtain the standard Pigouvian result that it is optimal to tax the externality-generating good at a constant rate equal to the marginal external damage associated with consuming the good (Kaplow, 2012).

2.1 Individual Maximization

Each agent maximizes her utility subject to her budget constraint and the tax system set by the social planner, producing indirect utility $v(w)$.

$$v(w) = \max_{z,x,c} u(c, x, z \mid w) \qquad (1)$$

$$\text{s.t.} \quad c + x + T_x(x) = z - T_z(z)$$

We assume that $u(c, x, z \mid w)$ is twice continuously differentiable, increasing in c and x, and decreasing in z. We further assume that $u(c, x, z \mid w)$ is strictly concave in c, x, and z.

We will often find it useful to approach (1) as a two-stage maximization problem. First, a type w chooses her income, $z(w)$. Second, she decides how to split the resulting disposable income between the two consumption goods, leading to choices $c(w)$ and $x(w)$. Specifically, a type w agent chooses taxable income:

$$z(w) \equiv \arg\max_z \left\{ u\left(z - T_z(z) - x(w; z) - T_x(x(w; z)), x(w; z), z \mid w\right) \right\} \qquad (2)$$

where:

$$x(w; z) \equiv \arg\max_x \left\{ u\left(z - T_z(z) - x - T_x(x), x, z \mid w\right) \right\} \qquad (3)$$

is the amount of the externality-generating good a type w agent would choose if her taxable income were z. The amount of the externality-generating good that a type w agent actually chooses is $x(w) \equiv x(w; z(w))$.

This approach allows us to write first-order conditions characterizing a type w agent's choices as a function of both tax schedules:

$$1 - T_z'(z) - \frac{\partial x(w;z)}{\partial z}\left(1 + T_x'(x)\right) = -\frac{u_z + \frac{\partial x(w;z)}{\partial z} u_x}{u_c} \tag{4}$$

$$\text{and} \qquad 1 + T_x'(x) = \frac{u_x}{u_c}. \tag{5}$$

For the sake of brevity we have suppressed the dependence of the marginal utility (e.g. u_c) terms on type (w) and/or on income (z). These two conditions can be combined to yield a reduced first order condition for the optimal choice of income: $1 - T_z'(z) = -\frac{u_z}{u_c}$, but this masks important intuition as we highlight below.

Equation 4 highlights something important: an agent's anticipation of her own behavior at stage two enters her first stage decision: For each additional dollar of income she considers earning, she evaluates how it would change her disposable income and how much she would spend on good x, given her preferences and the tax rate. In this way, the agent can be thought of as choosing her level of taxable income subject to an *effective net of income tax rate* of $1 - T_z'(z) - \frac{\partial x(w;z)}{\partial z} T_x'(x)$. Thus, when a rational agent sees her purchasing power decline due to commodity taxes, this reduces her marginal benefit of earning income. The size of this reduction depends on what fraction of her marginal dollar of disposable income will be spent on the dirty good. Critically for the analysis below, this marginal spending behavior may differ across people.

2.2 Social Welfare Maximization

A welfarist social planner chooses twice-differentiable tax functions, T_z and T_x. It does so to jointly maximize social welfare, $\boldsymbol{\mathcal{W}}$, while raising enough revenue to cover an exogenous revenue requirement, $\bar{\mathcal{R}}$.

$$\max \boldsymbol{\mathcal{W}} = \int \gamma(w) v(w) dF - \mathcal{D}(\overline{x}) \tag{6}$$

$$\text{s.t.} \quad \mathcal{R} \equiv \int \left(T_z(z(w)) + T_x(x(w))\right) dF(w) = \bar{\mathcal{R}} \tag{7}$$

$$\text{and} \quad \overline{x} = \int x(w) dF$$

Here, $\gamma(w)$ are Pareto weights; and $\mathcal{D}(\overline{x})$ is a separable atmospheric externality, which is an increasing function of aggregate consumption of the externality-generating good.[5]

2.3 Regularity Assumptions

For all tax schedules we consider, we assume that the optimal choices of c, x, and z are globally unique for every individual, with the relevant second-order conditions holding

[5] This is equivalent to having the harm from the externality-generating good entering individuals' utility in a separable way. To see the equivalence, consider $\tilde{u} \equiv u(c, x, z \mid w) - \widetilde{x}(\bar{x}; w)$, then $\tilde{v}(w) = v(w) - \widetilde{x}(\bar{x}; w)$ and $\mathcal{D}'(\overline{x}) = \int \gamma(w) \widetilde{x}'(\bar{x}; w) dF$. Our formulas are also locally unaffected if we define welfare using a social welfare function, $\mathcal{W} \equiv \int W(v(w)) dF - \mathcal{D}(\overline{x})$, rather than Pareto weights.

strictly. We also assume there is a smooth and strictly monotonic mapping from w to $c(w)$, $x(w)$, and $z(w)$. This is equivalent to the approach of Jacquet and Lehmann (2021) of making a single-crossing assumption and verifying monotonicity. These assumptions are standard and are required to ensure that choices move smoothly as the tax system changes, but imply restrictions on the set of tax schedules under consideration.

2.4 Characterizing Tax Reforms

We characterize the welfare effects of tax reforms using a formalization of the perturbation method of Saez (2001) via Gateaux derivatives, an approach discussed more thoroughly in Golosov et al. (2014), Jacquet and Lehmann (2021), and Gerritsen (2024). Given initial tax schedules $\mathcal{T}_x(x)$ and $\mathcal{T}_z(z)$, we ask how welfare and revenue change when we perturb them in directions $\tau_x(x)$ and $\tau_z(z)$ respectively. We assume that both $\tau_x(x)$ and $\tau_z(z)$ are twice continuously differentiable and compactly supported (or linear in subsection 2.11). Letting $\kappa \in \mathbb{R}$ index the magnitude of the reform, the new tax schedules are:

$$T_z(z;\kappa) \equiv \mathcal{T}_z(z) + \kappa\tau_z(z) \qquad \text{and} \qquad T_x(x;\kappa) \equiv \mathcal{T}_x(x) + \kappa\tau_x(x). \tag{8}$$

All our results about efficient and optimal tax policy will be based on examining the first-order effect of such perturbations on the social planner's Lagrangian,

$$\left.\frac{\partial\mathcal{L}}{\partial\kappa}\right|_{\kappa=0} = \frac{1}{\lambda}\left.\frac{\partial\mathcal{W}}{\partial\kappa}\right|_{\kappa=0} + \left.\frac{\partial\mathcal{R}}{\partial\kappa}\right|_{\kappa=0}, \tag{9}$$

starting from the initial tax system. Here, λ is the multiplier on the government's budget constraint. For any initial tax system, and proposed set of reforms, if $\frac{\partial\mathcal{L}}{\partial\kappa}\big|_{\kappa=0} > 0$, then movement in that direction is welfare-improving. At the optimal tax system, no such improvements can be possible and thus, at the optimum, we must have:

$$\left.\frac{\partial\mathcal{L}}{\partial\kappa}\right|_{\kappa=0} = 0 \quad \text{for all } \tau_z(\cdot) \text{ and } \tau_x(\cdot). \tag{10}$$

2.5 Distribution-Neutral Reforms

To begin our investigation of the optimal tax system, consider a special type of tax reform: one which is *distribution-neutral*.

Let $\hat{w}(z)$ be the type of agent that chooses income z. In other words, it is the inverse of the taxable income choice function, $z(w)$. This allows us to define $\hat{x}(z) \equiv x(\hat{w}(z);z)$, the dirty good consumption of an agent with taxable income z.

A distribution-neutral tax reform is one where for a given change in the dirty good tax liability of an agent with taxable income z, $\tau_x(\hat{x}(z))$, we also change her income tax liability by a precisely offsetting amount, such that the agent can still afford her original bundle. That is to say, the income tax reform is defined by:

$$\tau_z\left(z\right)=-\tau_x\left(\hat{x}\left(z\right)\right). \tag{11}$$

Notice, this implies marginal income tax rate changes of:

$$\tau_z'\left(z\right)=-\hat{x}'\left(z\right)\tau_x'\left(\hat{x}\left(z\right)\right), \tag{12}$$

where $\hat{x}'\left(z\right)$ is the slope of the observed cross-sectional relationship between dirty good consumption and taxable income.

Distribution-neutral reforms are of special interest because of the relative simplicity of their impact on social welfare. For any such reform, there is no first-order impact on agent utility, nor is there a mechanical impact on revenue. All effects of the reform arise because behavioral responses affect the externality and government revenue.

This also implies that for any distribution-neutral reform, if $\frac{\partial\mathcal{L}}{\partial\kappa}\big|_{\kappa=0}>0$, then a reform in that direction is a Pareto improvement: Positive gains in government revenue net of externality costs can be obtained without changes to individual agent utility, which allows a small transfer to be made that makes all agents strictly better off. Put another way, for any Pareto efficient tax system, it must be the case that:

$$\left.\frac{\partial\mathcal{L}}{\partial\kappa}\right|_{\kappa=0}=0\quad\text{for all }\tau_z(\cdot)\text{ and }\tau_x(\cdot)\text{ satisfying }\tau_z\left(z\right)=-\tau_x\left(\hat{x}\left(z\right)\right) \tag{13}$$

2.6 Behavioral Responses

By construction, any distribution-neutral reform produces a change in relative prices without a change in agents' purchasing power. Thus, only compensated behavioral responses will be relevant to the analysis. To establish what kinds of responses will occur, we can revisit the agent's first-order conditions.

Consumption of the Dirty Good (Direct). The reform to T_x changes the stage 2 first-order condition of an agent with taxable income z in the following way:

$$1+\mathcal{T}_x'\left(\hat{x}\left(z\right)\right)+\kappa\tau_x'\left(\hat{x}\left(z\right)\right)=\frac{u_x}{u_c}. \tag{14}$$

Holding constant the agent's choice of z, this generates a behavioral response of:

$$\left.\frac{\partial\hat{x}(z)}{\partial\kappa}\right|_{z,\,\kappa=0}=-\frac{\varepsilon_{x|z}(z)\,\hat{x}(z)}{1+\mathcal{T}_x'(\hat{x}(z))}\tau_x'(\hat{x}(z)) \tag{15}$$

where:

$$\varepsilon_{x|z}(z)\equiv-\frac{1+\mathcal{T}_x'(\hat{x}(z))}{\hat{x}(z)}\left.\frac{\partial\hat{x}(z)}{\partial\mathcal{T}_x'}\right|_z^c \tag{16}$$

is the compensated elasticity of spending on x with respect to its tax-inclusive price.

Income. Turning our attention to the stage 1 first-order condition, we have the following perturbed first-order condition:

$$1-\mathcal{T}_z'(z)+\underbrace{\kappa\hat{x}'(z)\,\tau_x'(\hat{x}(z))}_{\text{Reform to }T_z}-x_{\text{inc}}'(z)(1+\mathcal{T}_x'(x)+\underbrace{\kappa\tau_x'(\hat{x}(z))}_{\text{Reform to }T_x})=-\frac{u_z+x_{\text{inc}}'(z)u_x}{u_c}. \quad (17)$$

where $x_{\text{inc}}'(z)\equiv\frac{\partial x(\hat{w}(z);z)}{\partial z}$ is the causal effect of increasing the taxable income of a type $\hat{w}(z)$ agent on her choice of dirty good consumption.

Notice, in the two-stage formulation of the agent's problem, that this first-order condition depends on the reforms to both tax schedules. This is because, when responding to the reform at stage 1, the agent anticipates how the reform to the tax on the dirty good will impact her marginal return to earning income, given that some of her income will be spent on that good. After rearranging the left hand side of the equation, the agent's effective marginal return to earning income becomes:

$$1-\mathcal{T}_z'(z)-x_{\text{inc}}'(z)\left(1+\mathcal{T}_x'(x)\right)+\kappa\left(\hat{x}'(z)-x_{\text{inc}}'(z)\right)\tau_x'(\hat{x}(z)). \quad (18)$$

This highlights that heterogeneity in preferences leads to a distinction between the causal effect of additional income on dirty good consumption and the cross-sectional relationship between income and dirty good consumption. Following Ferey et al. (2024), we let $x_{\text{het}}'(z)$ be the component of the cross-sectional relationship that is driven by taste heterogeneity:

$$x_{\text{het}}'(z)\equiv\hat{x}'(z)-x_{\text{inc}}'(z). \quad (19)$$

As equation 17 shows, the change in the effective marginal income tax rate induced by the reform is directly proportional to this preference heterogeneity effect. As a result, the behavioral response of taxable income induced by the reform is:

$$\left.\frac{\partial z}{\partial\kappa}\right|_{\kappa=0}=\frac{z\varepsilon_z(z)}{1-\mathcal{T}_z'(z)}x_{\text{het}}'(z)\tau_x'(\hat{x}(z)) \quad (20)$$

where:

$$\varepsilon_z(z)\equiv\frac{1-\mathcal{T}_z'(z)}{z}\left.\frac{\partial z}{\partial\left(1-\mathcal{T}_z'(z)\right)}\right|_z^c \quad (21)$$

is the compensated elasticity of taxable income with respect to the net-of-income-tax rate. Note that formal definitions of these elasticities are included in the proof of Proposition 1.

Consumption of the Dirty Good (Indirect). The final behavioral response is the impact of the induced change in income on consumption of x. For an agent earning income z, this is determined by the causal effect of income on the dirty good,

$$\left.\frac{\partial \hat{x}(z)}{\partial z}\right|_{\kappa=0}\left.\frac{\partial z}{\partial \kappa}\right|_{\kappa=0}=x_{\text{inc}}^{\prime}(z)\frac{z\varepsilon_{z}(z)}{1-\mathcal{T}_{z}^{\prime}(z)}x_{\text{het}}^{\prime}(z)\tau_{x}^{\prime}(\hat{x}(z)), \tag{22}$$

resulting in a total effect of the reform on dirty good consumption of:

$$\left.\frac{\mathrm{d}x}{\mathrm{d}\kappa}\right|_{\kappa=0}=\left.\frac{\partial \hat{x}(z)}{\partial \kappa}\right|_{z,\,\kappa=0}+\left.\frac{\partial x}{\partial z}\right|_{\kappa=0}\left.\frac{\partial z}{\partial \kappa}\right|_{\kappa=0}. \tag{23}$$

2.7 A Key Sufficient Statistic: The "Taste Elasticity"

The formulation above highlights the key role of heterogeneity in preferences. Specifically, our optimal tax conditions below feature an important additional sufficient statistic: the "taste elasticity", $\eta_{x,z}^{\text{Taste}}(z)$. This is the elasticity of consumption of x with respect to income that occurs only through heterogeneity in agent tastes:

$$\eta_{x,z}^{\text{Taste}}(z)\equiv\frac{zx_{\text{het}}^{\prime}(z)}{\hat{x}(z)}. \tag{24}$$

The taste elasticity captures any differences in consumption of x across income that occur not due to the causal effect of higher income, but due to differences in the preferences of agents who locate at different parts of the income distribution.

In the Atkinson and Stiglitz (1976) benchmark in which preferences are identical across agents and separable from labor supply, $\eta_{x,z}^{\text{Taste}}(z)=0$. In this special case, both the income and indirect consumption effects (equations 20 and 22) are zero. The logic of the two-stage decision process above sheds light on why. Recall that a distribution-neutral reform uses the income tax to exactly offset the financial impact of the altered tax on the dirty good. If all agents have the same preferences, this also means that every agent's evaluation of every counterfactual income choice is also unchanged, because all agents would make the same consumption decisions if they had the same income level. This implies that the reform induces no change to any agent's choice of income.

More generally, differences in tastes across types can lead to a positive or negative value of $\eta_{x,z}^{\text{Taste}}(z)$, depending on whether higher or lower types have a relative preference for the dirty good. This implies potential labor supply impacts from the distribution-neutral reform. To see why, recall the two-stage decision process. An agent of type w_0 and income z_0 considers whether to increase her income slightly to z_1. If $\eta_{x,z}^{\text{Taste}}(z)>0$, she would consume less of x given income z_1 than type w_1 agents (who do have income z_1). Now consider a reform that increases $T_x(\hat{x}(z_1))$ while lowering $T_z(z_1)$ just enough to leave the utility of type w_1 unchanged at z_1. This same reform must raise the utility that w_0 would receive at z_1, making that higher income more attractive. Conversely, if $\eta_{x,z}^{\text{Taste}}(z)<0$, the same distribution-neutral reform reduces the incentive to increase one's income.

Another way of seeing this is that, if $\eta_{x,z}^{\text{Taste}}(z) = 0$, there is no "tagging" role for carbon in the sense of Akerlof (1978): Because tastes for carbon-intensive goods do not vary along the income distribution, carbon intensity does not reveal anything about a household's type that is not already revealed by her income choice. Only if $\eta_{x,z}^{\text{Taste}}(z) \neq 0$ will taxing or subsidizing carbon be useful for redistribution.

2.8 Testing Local Pareto Efficiency: A Specific Reform

To characterize the set of Pareto efficient tax systems, we proceed heuristically by considering a reform that raises the marginal commodity tax rate at a single income level. Our proofs in Appendix C provide a more formal treatment.[6] The reform raises the marginal commodity tax rate only for agents earning taxable income z, increasing it:

$$\tau_x'(\hat{x}(z')) = \begin{cases} 1 & \text{if } z' = z \\ 0 & \text{otherwise} \end{cases}. \tag{25}$$

The magnitude of the reform is carried by κ, as above. Distribution-neutrality then requires an offsetting reduction of those same agents' marginal income tax rates: by equation 12, $\tau_z'(z') = -\hat{x}'(z')$ if $z' = z$ and $\tau_z'(z') = 0$ otherwise.

2.9 Welfare Effects of the Reform

We obtain the welfare effect of a reform by multiplying each behavioral response (equations 15, 20, and 22) by its marginal impact on social welfare. For changes in consumption of the dirty good (equations 15 and 22), the marginal social value is the difference between the marginal tax rate on that good and its marginal social damage, i.e. $\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}$. For changes in income, the marginal social value is simply the income tax rate $\mathcal{T}_z'(z)$. In total, the local impact on social welfare at income z is the sum of three compensated revenue effects — with revenue from the dirty good valued net of its uninternalized damage:

$$\Delta\mathcal{R}(z) = \Delta\mathcal{R}^z(z) + \Delta\mathcal{R}^{x|z}(z) + \Delta\mathcal{R}^{z\to x}(z) \tag{26}$$

where the components are defined as follows.

$$\Delta\mathcal{R}^z(z) = \mathcal{T}_z'(z)\frac{z\varepsilon_z(z)}{1-\mathcal{T}_z'(z)}x_{\text{het}}'(z) \tag{27}$$

[6] In the appendix, we apply the fundamental lemma of the calculus of variations, similar to Jacquet and Lehmann (2021). Alternatively, our results can be formalized through a limiting argument—taking the point reform as the limit of reforms concentrated on a vanishing band around z—in the spirit of Gerritsen (2024).

$$\Delta\mathcal{R}^{x|z}(z) = -\left(\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)\frac{\varepsilon_{x|z}(z)\,\hat{x}(z)}{1+\mathcal{T}_x'(\hat{x}(z))} \tag{28}$$

$$\Delta\mathcal{R}^{z\rightarrow x}(z) = \left(\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)x_{\text{inc}}'(z)\frac{z\varepsilon_z(z)}{1-\mathcal{T}_z'(z)}x_{\text{het}}'(z) \tag{29}$$

The first of these effects, $\Delta\mathcal{R}^z$, captures the impact on government revenue from taxable income changes. The second, $\Delta\mathcal{R}^{x|z}$, is the uninternalized component of the externality caused by changes in consumption of the externality-generating good, which are directly caused by the tax reform. Finally, $\Delta\mathcal{R}^{z\rightarrow x}$ is the effect of further changes in consumption of x that are induced indirectly by changes in taxable income.

2.10 A Necessary Condition for Pareto Efficiency

Setting the local welfare effect $\Delta\mathcal{R}(z)$ to zero yields an optimal tax condition at each income level z: Because z was arbitrary, Pareto efficiency requires that there be no net gain from a distribution-neutral reform at any income.

$$\frac{\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1+\mathcal{T}_x'\left(\hat{x}\left(z\right)\right)} = \eta_{x,z}^{\text{Taste}}\left(z\right)\frac{\varepsilon_z\left(z\right)}{\varepsilon_{x|z}\left(z\right)}\left(\frac{\mathcal{T}_z'\left(z\right) + \left[\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right]x_{\text{inc}}'(z)}{1-\mathcal{T}_z'\left(z\right)}\right) \tag{30}$$

Our final step is to rearrange equation 30 to isolate the wedge between the marginal tax rate on x and its marginal damage, i.e. $\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}$. This yields Proposition 1.

Proposition 1. *If the income and commodity taxes are optimal and are allowed to be nonlinear, they must satisfy equation 31:*

$$\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda} = \frac{RR(z)\mathcal{T}_z'\left(z\right)}{1-RR(z)x_{inc}'(z)} \tag{31}$$

where:

$$RR(z) \equiv \eta_{x,z}^{Taste}\left(z\right)\frac{\varepsilon_z\left(z\right)}{\varepsilon_{x|z}\left(z\right)}\frac{1+\mathcal{T}_x'\left(\hat{x}\left(z\right)\right)}{1-\mathcal{T}_z'\left(z\right)} \tag{32}$$

measures relative responsiveness to changes in the carbon versus the income tax rates. If this condition is not satisfied, there exists a simultaneous reform to both taxes that yields a Pareto gain.

From Equation 31, we can see that $\eta_{x,z}^{\text{Taste}}\left(z\right)$ is a sufficient statistic that determines the sign and scales the size of any deviation from setting the marginal tax on x equal to the marginal damage, $\mathcal{D}'(\overline{x})/\lambda$. If there is no taste heterogeneity, i.e., $\eta_{x,z}^{\text{Taste}}\left(z\right) = 0$, the optimal corrective tax is the standard Pigouvian one: $\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) = \mathcal{D}'(\bar{x})/\lambda$. However, if $\eta_{x,z}^{\text{Taste}}\left(z\right) \neq 0$, the optimal carbon tax differs from the Pigouvian level. Any such deviations may vary across income levels, which could motivate means-tested rebates or targeted subsidies for externality-producing goods.

Unlike the standard Pigouvian tax, the optimal tax depends on the size of behavioral responses. The stronger the demand response (i.e. the larger is $\varepsilon_{x|z}$), the closer the optimal tax will be to the Pigouvian level. This happens because the planner's ability to successfully exploit tagging and use deviations from the Pigouvian level of the tax to redistribute depends on how responsive consumption of the tag is to taxation. A higher value of the elasticity of income (ε_z) favors a larger deviation from Pigouvian taxation: If the income tax is highly distortionary, using other instruments such as the carbon tax can be an efficient alternative. For a similar reason, we see that the deviation from Pigouvian taxation is proportional to the level of the marginal tax rate on income: A low income tax rate implies a small marginal distortion from the income tax in the first place.

The optimal tax could even have the opposite sign of the marginal damage. In this case, a good causing a negative externality is optimally subsidized, or a good causing a positive externality is taxed. This occurs when the taste elasticity is negative and the implied downward deviation exceeds the marginal damage itself, which is more likely when the externality is small or the relative responsiveness of income $\varepsilon_z/\varepsilon_{x|z}$ is large.

The denominator of equation 31 amplifies positive deviations from the Pigouvian level, and dampens negative deviations. Suppose that the taste elasticity is positive, motivating a carbon tax greater than the Pigouvian level. Raising the carbon tax and lowering the income tax produces efficiency gains. Incomes increase. If $x'_{\text{inc}}(z) > 0$, consumption of the carbon-intensive good increases as well. There is an uninternalized social *benefit* to that to the extent that $\mathcal{T}'_x\left(\hat{x}\left(z\right)\right) > \frac{\mathcal{D}'(\bar{x})}{\lambda}$, amplifying the original gains further. Conversely, suppose that the taste elasticity is negative. Then we realize efficiency gains as we lower the carbon tax below the Pigouvian level, the resulting rise in incomes and carbon-intensive consumption produces an uninternalized social *cost* because $\mathcal{T}'_x\left(\hat{x}\left(z\right)\right) < \frac{\mathcal{D}'(\bar{x})}{\lambda}$. This makes the efficiency gains from lowering the carbon tax less attractive than otherwise.

A different way of seeing this is that using the carbon tax for redistribution is more beneficial if the income tax is more distortionary. The net wedge from the income tax is:

$$\mathcal{T}'_z\left(z\right) + \left[\mathcal{T}'_x\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x'_{\text{inc}}(z). \tag{33}$$

Raising $\mathcal{T}'_x$ increases this wedge, making the income tax more distortionary and redistribution using the carbon tax more beneficial. This amplifies positive deviations from the Pigouvian level, while negative deviations are dampened.

A very similar approach can be used to derive conditions characterizing the optimal *levels* of the externality and income taxes; we discuss these conditions in Appendix B.

2.11 Restricting to Linear Taxation

Nonlinear pricing of carbon-intensive goods has real-world precedents for goods such as electricity. However, this may not be true in all settings or for all externality-generating goods. We therefore also consider the optimal linear tax and the extent to which this deviates from the level of marginal damage.

If the tax on the externality-generating good must be linear ($\mathcal{T}_x'(\hat{x}(z)) = t_x$), the Pareto efficiency condition for the linear tax rate is similar to the condition for the nonlinear corrective tax (Equation 31). However, it averages over all levels of income z and consumption of x. Specifically, the optimal linear tax is characterized by Proposition 2.

Proposition 2. *If the income and commodity taxes are optimal and the corrective tax is restricted to being linear, they must satisfy equation 34:*

$$t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda} = \frac{\mathbb{E}_z\left[RR(z)\,\mathcal{T}_z'(z)\right]}{1 - \mathbb{E}_z\left[RR(z)\,x'_{inc}(z)\right]} \tag{34}$$

where $\mathbb{E}_z$ is the expectation over the income distribution, and:

$$RR(z) \equiv \eta_{x,z}^{Taste}(z)\,\frac{\varepsilon_z(z)\,\hat{x}(z)}{\mathbb{E}_z\left[\varepsilon_{x|z}(z)\,\hat{x}(z)\right]}\left(\frac{1+t_x}{1-\mathcal{T}_z'(z)}\right). \tag{35}$$

and $x'_{inc}(z) = \frac{\partial x(\hat{w}(z);z)}{\partial z}$ is the causal effect of income on consumption of x. If this condition is not satisfied, there exists a simultaneous reform to both taxes that yields a Pareto gain.

The derivation is straightforward. For a linear dirty good tax, the reforms we can consider are restricted to $\tau_x(\hat{x}(z)) \equiv \hat{x}(z)$. Plugging this into the equation for the welfare impact (equation 26) and setting it to zero gives us equation 34.

3 Multidimensional Heterogeneity

In practice, households with the same income may differ in their preferences for carbon-intensive goods due to factors such as geographic location, household composition, or idiosyncratic tastes.[7] We abstracted from this type of heterogeneity in our unidimensional model, but now extend the model to allow for it.

Within-income heterogeneity makes exact compensation through the income tax impossible: When households with the same income consume different amounts of the carbon-intensive good, any income-based adjustment will give some households too much compensation and others too little. Consequently, we consider the closest available analogue of the distribution-neutral reforms of Section 2: *vertically neutral* reforms, which

[7] Within-income heterogeneity has been studied empirically by Pizer and Sexton (2019), Cronin et al. (2019), and Rausch et al. (2011). Sallee (2022) discusses the potential for true Pareto gains in that case.

leave the average tax liability at each level of income unchanged. Vertically neutral reforms eliminate mechanical redistribution between income levels, but not within them.

Our results provide a characterization of the optimal tax system that is broadly similar to that of Section 2, but augmented in several ways to account for imperfect compensation. We show that, all else equal, within-income heterogeneity in x'_{inc} results in an attenuation of the optimal deviation from the Pigouvian benchmark.

To simplify the discussion, we focus on the case of linear commodity taxation. To adapt the model to incorporate multidimensional heterogeneity, let each agent have type $(w, \theta) \in \mathbb{R}_{++} \times \Theta$. Type (w, θ) agents make choices to maximize their utility $u(c, x, z; w, \theta)$. We largely maintain the assumptions of the unidimensional case in Section 2.3, which ensure that choices of $z(w, \theta)$ and $x(w, \theta)$ are unique and respond smoothly to reforms. In addition, consistent with Jacquet and Lehmann (2021), we assume that the mapping $w \mapsto z(w, \theta)$ is strictly monotone. This ensures that the conditional inverse, $\hat{w}(z; \theta)$, is well-defined. We further assume that, for all z, the conditional mean, $\bar{x}(z) \equiv \mathbb{E}[x(w, \theta)|z(w, \theta) = z]$, exists and is continuously differentiable.

3.1 Vertically Neutral Reforms

Formally, we define a vertically neutral tax reform as a pair of reform functions τ_x and τ_z such that, for each income level z:

$$\tau_z(z) = -\mathbb{E}[\tau_x(x(w, \theta))|z(w, \theta) = z]. \tag{36}$$

In the linear tax case, this is simply $\tau_z(z) = -\bar{x}(z)$, with an implied reform to marginal income tax rates of $\tau'_z(z) = -\bar{x}'(z)$.

The welfare effect of a vertically neutral reform is:

$$\left.\frac{\partial \mathcal{L}}{\partial \kappa}\right|_{\kappa=0} = \underbrace{\int \mathbb{E}\left[(1 - g(z; \theta)) \cdot (\hat{x}(z, \theta) - \bar{x}(z)) \mid z\right] h_z(z)\, dz}_{\text{Welfare effect from mechanical redistribution}} + \underbrace{\int \left[\overline{\Delta\mathcal{R}}^{x|z}(z) + \overline{\Delta\mathcal{R}}^{z}(z) + \overline{\Delta\mathcal{R}}^{z \to x}(z)\right] h_z(z)\, dz}_{\text{Revenue effects from compensated behavioral responses}}, \tag{37}$$

where $g(z; \theta)$ is the marginal social welfare weight of a type θ household with income z, and $\hat{x}(z, \theta) \equiv x(\hat{w}(z; \theta), \theta)$ is its consumption of the dirty good.

The first term is the direct consequence of imperfect compensation, and it has no counterpart in Section 2. Full compensation would require adjusting each household's tax liability in line with its own consumption of the dirty good. Instead, a vertically neutral reform adjusts it in line with average consumption at the household's income level. The discrepancy between the two is a change in tax liability of $(\hat{x}(z, \theta) - \bar{x}(z))\, \kappa$. Households that consume more than average at their income level are undercompensated, and households that consume less are overcompensated. The first term values this within-income redistribution using the welfare weights. Equivalently, it equals $-\int \mathbb{C}\left[g(z;\theta), \hat{x}(z,\theta) \mid z\right] h_z(z)\, dz$: what matters is not within-income dispersion in consumption per se, but whether that dispersion covaries with the welfare weights.

The second term in equation 37 collects the same three compensated behavioral responses as in Section 2.6, averaged over the types present at each income level:

$$\overline{\Delta\mathcal{R}}^{x|z}(z) \equiv \mathbb{E}\left[-\left(t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)\frac{\varepsilon_{x|z}(z;\theta)\hat{x}(z,\theta)}{1+t_x} \,\middle|\, z\right] \tag{38}$$

$$\overline{\Delta\mathcal{R}}^{z}(z) \equiv \mathbb{E}\left[\mathcal{T}_z'(z)\frac{\varepsilon_z(z;\theta)z}{1-\mathcal{T}_z'(z)}x'_{\text{het}}(z;\theta) \,\middle|\, z\right] \tag{39}$$

$$\overline{\Delta\mathcal{R}}^{z\to x}(z) \equiv \mathbb{E}\left[\left(t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)x'_{\text{inc}}(z;\theta)\frac{\varepsilon_z(z;\theta)z}{1-\mathcal{T}_z'(z)}x'_{\text{het}}(z;\theta) \,\middle|\, z\right] \tag{40}$$

As before, $\overline{\Delta\mathcal{R}}^{x|z}$ captures changes in consumption of x directly caused by the reform to t_x; $\overline{\Delta\mathcal{R}}^{z}$ captures the impact on government revenue from taxable income choices; and $\overline{\Delta\mathcal{R}}^{z\to x}$ is the effect of further changes in consumption of x induced indirectly by changes in income. The objects inside each average are those of Section 2.6, indexed by θ. In particular, the elasticities $\varepsilon_{x|z}(z;\theta)$ and $\varepsilon_z(z;\theta)$ and the causal income effect $x'_{\text{inc}}(z;\theta)$ may now vary across households with the same income.

The discrepancy between required and provided compensation operates on the behavioral responses as well. The income tax offset adjusts marginal income tax rates in line with average consumption, by $-\bar{x}'(z)\kappa$, while the change in a household's marginal value of income depends on its own consumption response, $x'_{\text{inc}}(z;\theta)$. The difference,

$$x'_{\text{het}}(z;\theta) \equiv \bar{x}'\left(z\right) - x'_{\text{inc}}(z;\theta), \tag{41}$$

is the type-specific analogue of the taste heterogeneity component of Section 2.6. A type θ agent with income z sees her effective marginal income tax rate change by $-\kappa x'_{\text{het}}(z;\theta)$, and her taxable income response is proportional to it — and with it, so is the induced indirect change in her consumption of the dirty good.

3.2 Heterogeneity in Behavioral Responses

Because the statistics inside each conditional average vary jointly with θ, the averaged behavioral responses depend not only on the average value of each statistic at a given income level, but also on the covariances between them.

Consider $\overline{\Delta\mathcal{R}}^{z}(z)$, the income tax revenue that results from the combined reform to $\mathrm{d}t_x$ and the income tax. The reform reduces a type θ agent's effective tax rate on additional income by $x'_{\text{het}}(z;\theta)\mathrm{d}t_x$. This varies across agents to the extent that there is heterogeneity in the causal income effect, $x'_{\text{inc}}(z;\theta)$. Then the response to that change varies with the elasticity of taxable income, $\varepsilon_z(z;\theta)$. We therefore cannot simply replace each behavioral parameter with its mean. Instead, we can write:

$$\overline{\Delta\mathcal{R}}^z(z) = \mathcal{T}'_z(z)\cdot\frac{\bar{\varepsilon}_z(z)z}{1-\mathcal{T}'_z(z)}\cdot\bar{x}'_{\text{het}}(z) + \frac{\mathcal{T}'_z(z)z}{1-\mathcal{T}'_z(z)}\mathbb{C}\left[\varepsilon_z(z;\theta), x'_{\text{het}}(z;\theta)|z\right]. \tag{42}$$

where $\bar{x}'_{\text{het}}(z)$ and $\bar{\varepsilon}_z(z)$ are the average values of $x'_{\text{het}}(z;\theta)$ and $\varepsilon_z(z;\theta)$ for agents at income level z, and $\mathbb{C}$ indicates a covariance. Using only the first term would amount to applying the unidimensional formula to average statistics. The covariance is a correction that takes into account within-income heterogeneity.

Turning to the indirect consumption effect, $\overline{\Delta\mathcal{R}}^{z\to x}(z)$, we can again rewrite this as a function of the average values of each statistic and their covariances:

$$\begin{aligned}\overline{\Delta\mathcal{R}}^{z\to x}(z) = &\left(\frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1-\mathcal{T}'_z(z)}\right) z\bar{\varepsilon}_z(z)\bar{x}'_{\text{inc}}(z)\bar{x}'_{\text{het}}(z) \\ &+ \left(\frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1-\mathcal{T}'_z(z)}\right) z\bar{\varepsilon}_z(z)\mathbb{C}\left[x'_{\text{inc}}(z;\theta), x'_{\text{het}}(z;\theta)|z\right] \\ &+ \left(\frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1-\mathcal{T}'_z(z)}\right) z\,\mathbb{C}\left[\varepsilon_z(z;\theta), x'_{\text{inc}}(z;\theta)x'_{\text{het}}(z;\theta)|z\right].\end{aligned} \tag{43}$$

The first term in this expression is simply the product of mean behavioral responses, which would also be present in a model without multidimensional heterogeneity.

The second and third terms again involve covariances. These as arise from the two steps that lead from the tax reform to indirect changes in consumption of x. First, the income response for an individual of type (w,θ) is proportional to $\varepsilon_z(z;\theta)\times x'_{\text{het}}(z;\theta)$. Second, fixing this income response, the income effect on consumption of x is proportional to $x'_{\text{inc}}(z;\theta)$. The definition of $x'_{\text{het}}(z;\theta)$ in terms of $x'_{\text{inc}}(z;\theta)$ means that the magnitudes of these forces are *mechanically* related at the individual level. As a result, the first covariance collapses to the negative variance of causal income effects at z:

$$\mathbb{C}\left[x'_{\text{inc}}(z;\theta), x'_{\text{het}}(z;\theta)\mid z\right] = -\mathbb{V}\left[x'_{\text{inc}}(z;\theta)\mid z\right]. \tag{44}$$

Intuitively, agents with a larger $x'_{\text{het}}(z;\theta)$ experience a greater than average change in their effective marginal income tax rates and those same agents have a smaller consumption response to any given increase in their incomes compared to the average.

The second covariance makes a further adjustment to account for the possibility that the product of the causal income and heterogeneity effects $x'_{\text{inc}}(z;\theta)x'_{\text{het}}(z;\theta)$ may be correlated with the elasticity of taxable income, $\varepsilon_z(z;\theta)$.

This highlights a useful and tractable special case: If we assume that $\varepsilon_z(z;\theta)$ is constant within income levels, then the third term is zero. However, the first covariance remains even then unless there is no within-income heterogeneity at all. In conclusion, multidimensional heterogeneity in the sense of variation in $x'_{\text{inc}}(z;\theta)$ across individuals with the same income leads to robust attenuation of the indirect consumption effect.

3.3 Optimal Linear Taxation: The Multidimensional Case

Our final step is to put together the impacts on social welfare from a vertically neutral reform, and present an optimal tax condition, harnessing the fact that the welfare effect of any vertically neutral reform (equation 37) must equal zero at the optimum.

By construction, the reforms we have considered do not redistribute vertically. For simplicity, we also assume throughout the analysis below that welfare weights are uncorrelated with consumption of the externality-generating good conditional on income, at least on average. That is, we assume that:

$$\mathbb{E}\left[\mathbb{C}\left(g\left(w,\theta\right),x\left(w,\theta\right)|z\left(w,\theta\right)=z\right)\right]=0. \tag{45}$$

In practice, the sign of this covariance depends on whether more intense consumers of carbon-intensive goods have higher or lower welfare weights conditional on income. The assumption is exactly the condition under which the mechanical-redistribution term in equation 37 vanishes. It makes the multidimensional condition empirically implementable at the cost of abstracting from differences in levels well-being between households with the same income, which are potentially relevant if correlated with carbon intensity.

Assuming that the effect on welfare of redistribution within income levels is negligible on average, rearranging equation 37 yields the following characterization of the optimal linear tax on the externality-generating good.

Proposition 3. *Assume that welfare weights are uncorrelated with consumption of the externality-generating good conditional on income:*

$$\mathbb{E}\left[\mathbb{C}(g(z;\theta),x(z;\theta)\mid z)\right]=0. \tag{46}$$

Then the optimal linear tax on the externality-generating good satisfies:

$$t_x-\frac{\mathcal{D}'(\bar{x})}{\lambda}=\frac{\mathbb{E}_z\left[\mathbb{E}_\theta\left[RR(z;\theta)\mathcal{T}_z'(z)\mid z\right]\right]}{1-\mathbb{E}_z\left[\mathbb{E}_\theta\left[RR(z;\theta)x'_{inc}(z;\theta)\mid z\right]\right]}, \tag{47}$$

where:

$$RR(z;\theta)\equiv\eta^{Taste}_{x,z}(z;\theta)\left(\frac{\varepsilon_z(z;\theta)\bar{x}(z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\right)\left(\frac{1+t_x}{1-\mathcal{T}_z'(z)}\right), \tag{48}$$

and the type-specific taste elasticity is:

$$\eta_{x,z}^{\textit{Taste}}(z;\theta) \equiv \frac{z x_{\textit{het}}'(z;\theta)}{\bar{x}(z)} = \frac{z(\bar{x}'(z) - x_{\textit{inc}}'(z;\theta))}{\bar{x}(z)}. \tag{49}$$

Equation (47) closely parallels the optimal linear tax condition from the unidimensional case (Proposition 2), but with expectations taken over both income levels and θ types within each income level. The taste elasticity $\eta_{x,z}^{\text{Taste}}(z;\theta)$ now measures how a type θ individual's causal income effect diverges from the overall cross-sectional relationship between income and consumption.

3.4 Decomposing the Multidimensional Optimal Tax Condition

To guide our empirical implementation, we can decompose the expectations in equation 47 into average effects and within-income covariance terms. This follows the same logic as when we decomposed the averaged behavioral responses in Section 3.2.

First, the numerator of the right-hand side of equation 47 can be written as the expectation of $\frac{(1+t_x)\,\mathcal{T}_z'(z)}{1-\mathcal{T}_z'(z)}\,\mathbf{N}(z)$ where:

$$\mathbf{N}(z) = \bar{\eta}_{x,z}^{\text{Taste}}(z)\frac{\bar{x}(z)\bar{\varepsilon}_z(z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]} - \frac{z\,\mathbb{C}(\varepsilon_z(z;\theta), x_{\text{inc}}'(z;\theta) \mid z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}.$$

Here, $\bar{\eta}_{x,z}^{\text{Taste}}(z)$ and $\bar{\varepsilon}_z(z)$ are averages across θ types at income z.

The first term of this equation involves average values of each sufficient statistic at each income level. To use this first term while ignoring the second would be equivalent to naively applying the unidimensional formula with the average values of each sufficient statistic. The second term is a correction involving the within-income covariance between the elasticity of taxable income and the causal income effect on x.

Consider a population where, conditional on z, agents with higher ε_z also have higher x_{inc}' (and thus lower $\eta_{x,z}^{\text{Taste}}$). Such agents have two key characteristics. First, they respond strongly to income tax changes. Second, the tagging value of carbon is low for such agents. Thus, if the planner could condition taxes on θ, they would set a smaller deviation from Pigouvian taxation for these agents. In reality, the planner can only condition on z. However, the optimal uniform policy must account for the fact that the most responsive agents are precisely those for whom tagging is least valuable.

Second, the denominator of the right-hand side of equation 47 can be written as $1 - \mathbb{E}_z\big[\frac{1+t_x}{1-\mathcal{T}_z'(z)}\,\mathbf{D}(z)\big]$ where:

$$\mathbf{D}(z) = \bar{\eta}_{x,z}^{\text{Taste}}(z)\,\frac{\bar{x}(z)\,\mathbb{E}[x_{\text{inc}}'(z;\theta)\,\varepsilon_z(z;\theta) \mid z]}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)\,x(z;\theta)]} - \frac{z\,\mathbb{C}\left(x_{\text{inc}}'(z;\theta),\, x_{\text{inc}}'(z;\theta)\,\varepsilon_z(z;\theta) \mid z\right)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)\,x(z;\theta)]}.$$

This covariance reflects correlation between the tagging value of the externality-generating good and indirectly induced consumption of that good. Suppose that among agents at a given level of income, two things go together: (1) a high tagging value of x, as reflected by a low value of x'_{inc}; and (2) a small indirect response of x because x'_{inc} is small, ε_z is small, or both. Then this covariance term will be positive, which increases the denominator and dampens any optimal deviation from the Pigouvian level of the optimal tax. As Section 3.2 highlights, there is a mechanical component to this because x'_{inc} enters both steps (1) and (2). Intuitively, the planner must take into account that agents for whom tagging is most valuable are precisely those for whom any efficiency gains from a reform lead indirectly to more counterproductive externality-generating consumption.

3.5 A Tractable Case: Constant Elasticity of Taxable Income

In our empirical application, we assume that the income elasticity does not vary with θ at each income level. Corollary 1 presents the optimal tax condition for this empirically implementable case. As expected from Section 3.2, the condition features the variance of x'_{inc}. Holding all other empirical objects constant, this extra term attenuates any deviation of the optimal tax on x from the Pigouvian level. Only in the special case in which there is no within-income heterogeneity, so that $\text{Var}(x'_{\text{inc}}(z;\theta) \mid z) = 0$, does equation (50) simplify to the unidimensional optimal linear tax condition in Proposition 2.

Corollary 1. If $\varepsilon_z(z;\theta) = \bar{\varepsilon}_z(z)$ for all θ at each income level z, the optimal linear tax satisfies:

$$t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda} = \frac{\mathbb{E}_z\left[\overline{\text{RR}}(z)\mathcal{T}'_z(z)\right]}{1 - \mathbb{E}_z\left[\overline{\text{RR}}(z)\bar{x}'_{\text{inc}}(z)\right] + \mathbb{E}_z\left[\widetilde{\text{RR}}(z)\mathbb{V}(x'_{\text{inc}}(z;\theta) \mid z)\right]}, \tag{50}$$

where:

$$\overline{\text{RR}}(z) \equiv \bar{\eta}^{\text{Taste}}_{x,z}(z)\left(\frac{\bar{\varepsilon}_z(z)\bar{x}(z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\right)\left(\frac{1+t_x}{1-\mathcal{T}'_z(z)}\right), \tag{51}$$

$$\widetilde{\text{RR}}(z) \equiv \frac{\bar{\varepsilon}_z(z)z}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\frac{1+t_x}{1-\mathcal{T}'_z(z)}, \tag{52}$$

and $\bar{x}'_{\text{inc}}(z) \equiv \mathbb{E}[x'_{\text{inc}}(z;\theta) \mid z]$ is the average causal income effect at z.

Corollary 1 identifies the sufficient statistics that are needed to characterize optimal policy with multidimensional heterogeneity, subject to our restriction that the elasticity of taxable income is constant. First, we require the average value of several statistics at each income level: the cross-sectional slope, $\bar{x}'(z)$; the causal income effect, $\bar{x}'_{\text{inc}}(z)$; the elasticity of taxable income, $\bar{\varepsilon}_z(z)$; and the weighted average demand elasticity, $\varepsilon_{x|z}(z;\theta)$. Second, we need to measure the within-income variance in income effects, $\text{Var}(x'_{\text{inc}}(z;\theta) \mid z)$.

4 Empirical Application

Equations 30, 34, and 50 allow us to quantify the optimal carbon tax for the United States while taking into account the distributional impacts. For a given marginal social cost of carbon, we show how the Pareto-efficient carbon tax may deviate from the standard Pigouvian correction under both nonlinear and linear commodity tax systems, presenting estimates for a range of different assumptions regarding the compensated elasticities of taxable income and dirty good consumption. We initially consider the simpler unidimensional case, and then we incorporate multidimensional heterogeneity.

We start with our empirical quantification of taste heterogeneity, $\eta_{x,z}^{\text{Taste}}(z)$. This is the focus of Section 4.1. In Section 4.2, we then describe how we estimate and calibrate the other key objects which jointly determine the optimal carbon tax.

4.1 Quantifying Taste Heterogeneity

To estimate the extent of taste heterogeneity, we use the decomposition of the cross-sectional relationship provided by equation 19. This requires an estimate of $\hat{x}'(z)$, which we obtain from cross-sectional variation in the consumer expenditure survey; and an estimate of x'_{inc} at each of many different points in the income distribution, which we obtain by running our own survey.

4.1.1 Measuring Cross-Sectional Variation Using The CEX

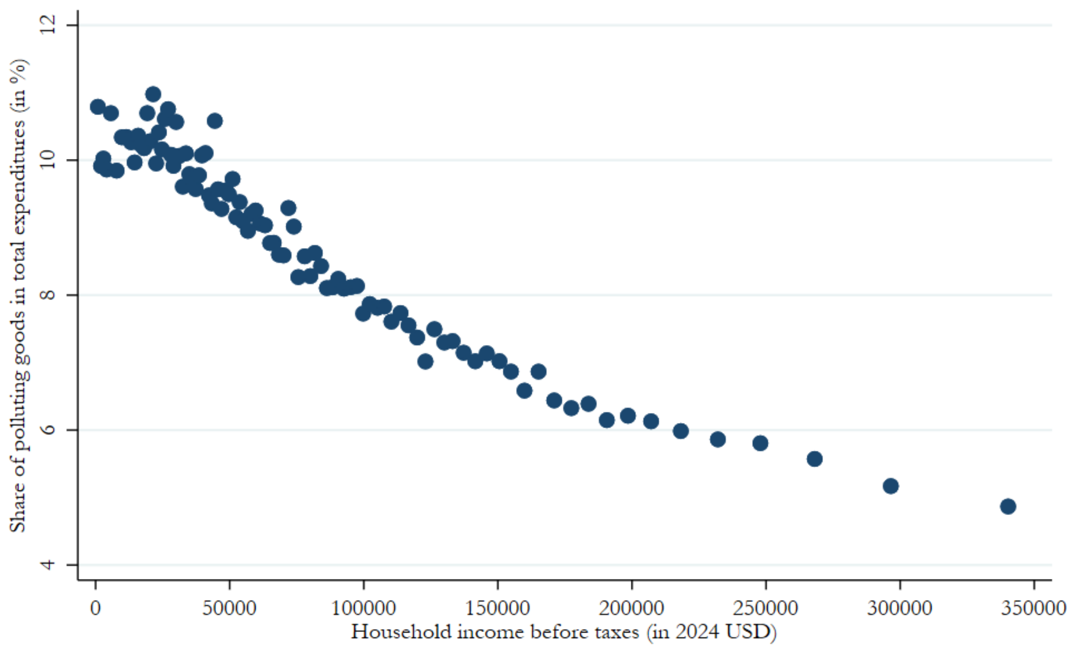
We measure the cross-sectional relationship between income and consumption of carbon-intensive goods ($\hat{x}'(z)$) using the Consumer Expenditure Surveys (CEX). Specifically, we use information from the public use microdata (PUMD) files of the 2018–2024 CEX surveys from the United States Bureau of Labor Statistics. We rely on the Interview survey, a rotating panel survey with approximately 6,000 observations per quarter, providing detailed data on expenditures, income, and characteristics of consumers. Our sample is composed of 112,885 quarterly observations across 47,054 households.[8] Because of the rotating nature of the panel, households may be sampled between one and four consecutive times (quarters) across the sample period (2018–2024).[9]

We compute expenditure on carbon-intensive goods as a percentage of total consumption for each quarterly household observation. We focus on two carbon-intensive expenditure categories: (i) an aggregate dirty good, and (ii) gasoline and other motor oils. Aggregate dirty good consumption is defined as the sum of expenditure on electricity, gasoline and motor oil, heating fuels, and natural gas. Appendix A.1 shows the share of consumption accounted for by these and other categories.

[8]See detailed definition of a "consumer unit" at: https://www.bls.gov/cex/csxfaqs.htm#qc1.

[9]For 2024, we use only Q1 data because tax variables are unavailable in later quarters.

(a) Aggregate Dirty Good



(b) Gasoline

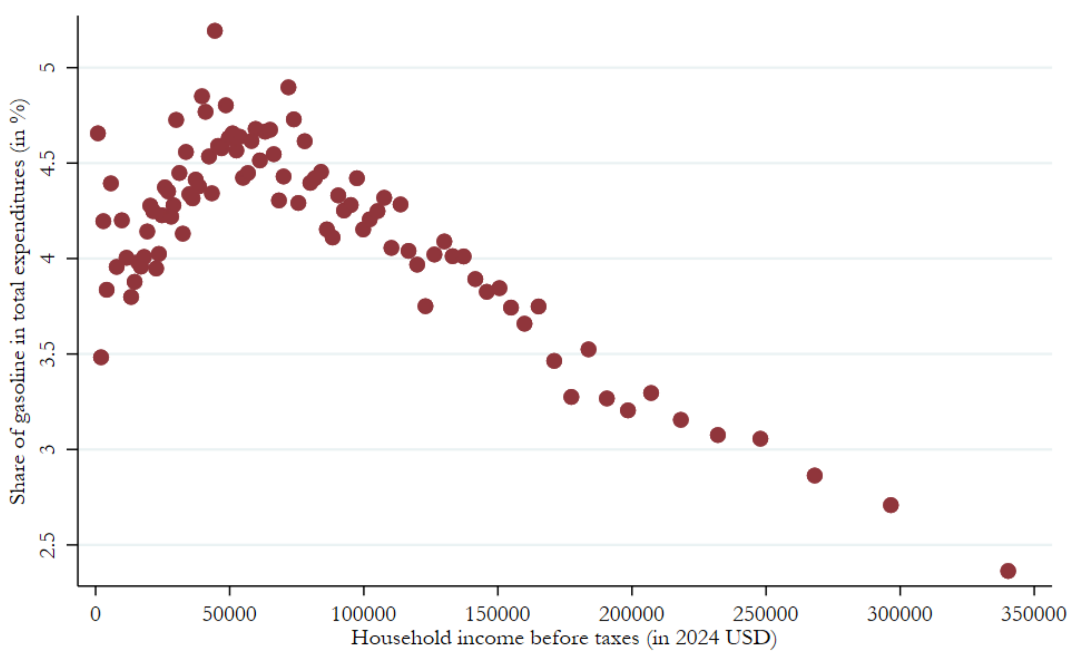


Figure 1. Expenditure shares of carbon-intensive goods across the income distribution

Note: These binscatter plots show (i) the average share (in %) of expenditure on polluting goods as a fraction of total consumption, and (ii) the corresponding average household taxable income, for each income percentile using data from the 2018–2024 Consumer Expenditure Survey quarterly waves. The "dirty good" expenditure category includes gasoline, electricity, natural gas, and heating fuels. Household taxable incomes were converted to 2024 USD using the average annual Consumer Price Index in the US. The figure is truncated at $350k for readability. This excludes the top 3% of earners.

To smooth the cross-sectional relationship between household income and consumption of carbon-intensive goods, we assign each household to a year-specific income percentile (because taxable income is measured annually).[10] We then average consumption and income across quarterly observations of households within each income percentile bin, using CEX-provided survey weights to ensure the sample is representative of the US population. This binning procedure abstracts away from any within-income heterogeneity in dirty good consumption. In Section 4.3.3, we relax this restriction.

Panel (a) of Figure 1 shows the average *dirty good* expenditure as a share (in percent) of total expenditures at each point in the income distribution. Panel (b) shows the same measure for *gasoline* only.[11] Poorer households spend a much larger share of their consumption budget on carbon-intensive goods than do higher-income households. For the aggregate dirty good displayed in panel (a), the share ranges from a high of 11.0% at the bottom of the income distribution to a low of approximately 3.8% at the very top. Focusing on gasoline only, displayed in panel (b), the expenditure share ranges from 5.2% at the bottom of the income distribution to 1.7% at the very top.

While consumption *shares* of carbon-intensive goods are generally decreasing with income (see Figure 1), it is important to note that *levels* are increasing with income. Appendix Figure A.1 shows the cross-sectional relationship between quarterly consumption expenditure (in dollars) of carbon-intensive goods and household taxable income. The slope is positive but flattens out at higher levels of household taxable income.

4.1.2 Measuring Income Effects Using A Hypothetical Choice Survey

Next, we measure the causal response of additional income, x'_{inc}. This requires a survey of our own, which measures how each household's consumption would respond to a small amount of additional income. We designed and ran the survey using *Qualtrics*, and recruited participants on *Prolific Academic*.[12] The sample comprises 1,448 US residents aged 25 to 54, and was designed to be representative of the US population along the targeted dimensions of age, gender, and race (see Appendix Table A.6).[13]

The primary purpose of the survey was to elicit how each household would respond to an unexpected increase in household (after-tax) income. Respondents were asked to

[10]We further decompose the top 1% of earners, into a 99.9th percentile bin, corresponding to the top 0.1% of CEX respondents, and leading to a total of 101 percentile bins.

[11]Using total expenditure rather than income as the denominator is preferable when studying the distributional burden of excise taxes because expenditures serve as a better proxy for lifetime income (Poterba, 1989). Using expenditures has the additional benefit of abstracting away from differences in saving rates across incomes. The main takeaway from Figure 1 remains the same if we compute shares of income.

[12]The survey and our analysis of the results were pre-registered on the Open Science Framework (OSF) platform in January 2025: `https://osf.io/swqze`.

[13]The survey was conducted in two waves, the first in February 2025, and the second in July 2025.

allocate a $1,000 permanent increase in household income across an exhaustive list of expenditure categories (mirroring those from the CEX), and a residual savings bucket.[14] This elicitation procedure provides us with each respondent's marginal propensity to consume (MPC) out of a permanent increase in household income, and its breakdown between carbon-intensive goods (electricity, natural gas, heating fuels, and gasoline) and other goods. We also collected information about respondents' *status quo* household income (both before- and after-tax), and the breakdown of these expenditures.

Response Quality. We took several steps to ensure the quality of respondents, including screening out respondents based on (i) past studies' rejection rates, and (ii) the number of hours spent on the platform. As stated in our pre-registration plan, we also filter out participants who fail attention checks, speedsters, extreme outliers, and respondents who provide inconsistent or nonsensical information on before- vs. after-tax income or marginal propensity to consume (see Appendix Table A.7 for a breakdown of sample exclusions).

In addition, we have a powerful way to validate the representativeness of our survey sample. Figure A.2 plots the relationship between our respondents' reported consumption expenditures and their incomes. This relationship can be compared to that in the Consumer Expenditure Survey (Figure 1). The figures align closely, suggesting that our responses are of high quality and our respondents quite representative.

Marginal Consumption Patterns In The Survey. Figure 2 shows a binscatter plot of the relationship between the share of carbon-intensive goods in total marginal consumption, and household taxable income. We also include corresponding global polynomial fits (of order 2). Panel (a) shows the aggregate results for all of the carbon-intensive goods, while panel (b) shows the results for gasoline. In both cases, we see a declining marginal propensity to consume these goods as income rises.

Estimating Causal Income Effects. Our estimates for x'_{inc} build on our survey respondents' marginal consumption choices in response to a hypothetical increase in their incomes (see Figure 2). However, we need to make a few additional assumptions to estimate $x'_{\text{inc}}(z)$ based on the consumption response to a small permanent increase in after-tax income.

First, note that in our theoretical framework, the response of x to a pure income shock (the quantity we observe in our custom survey) is given by:

$$\frac{dx\left(\hat{w}\left(z\right);z\right)}{dI} = \eta_{x|z}(z) + \frac{\partial x\left(\hat{w}\left(z\right);z\right)}{\partial z}\,\eta_z(z), \tag{53}$$

where η_z and $\eta_{x|z}$ are income effects due to a reform that lowers the tax burden slightly, while leaving both marginal tax rates unchanged (see Appendix C for formal definitions).

[14] See Figure D.1 for a screenshot of the online survey page.

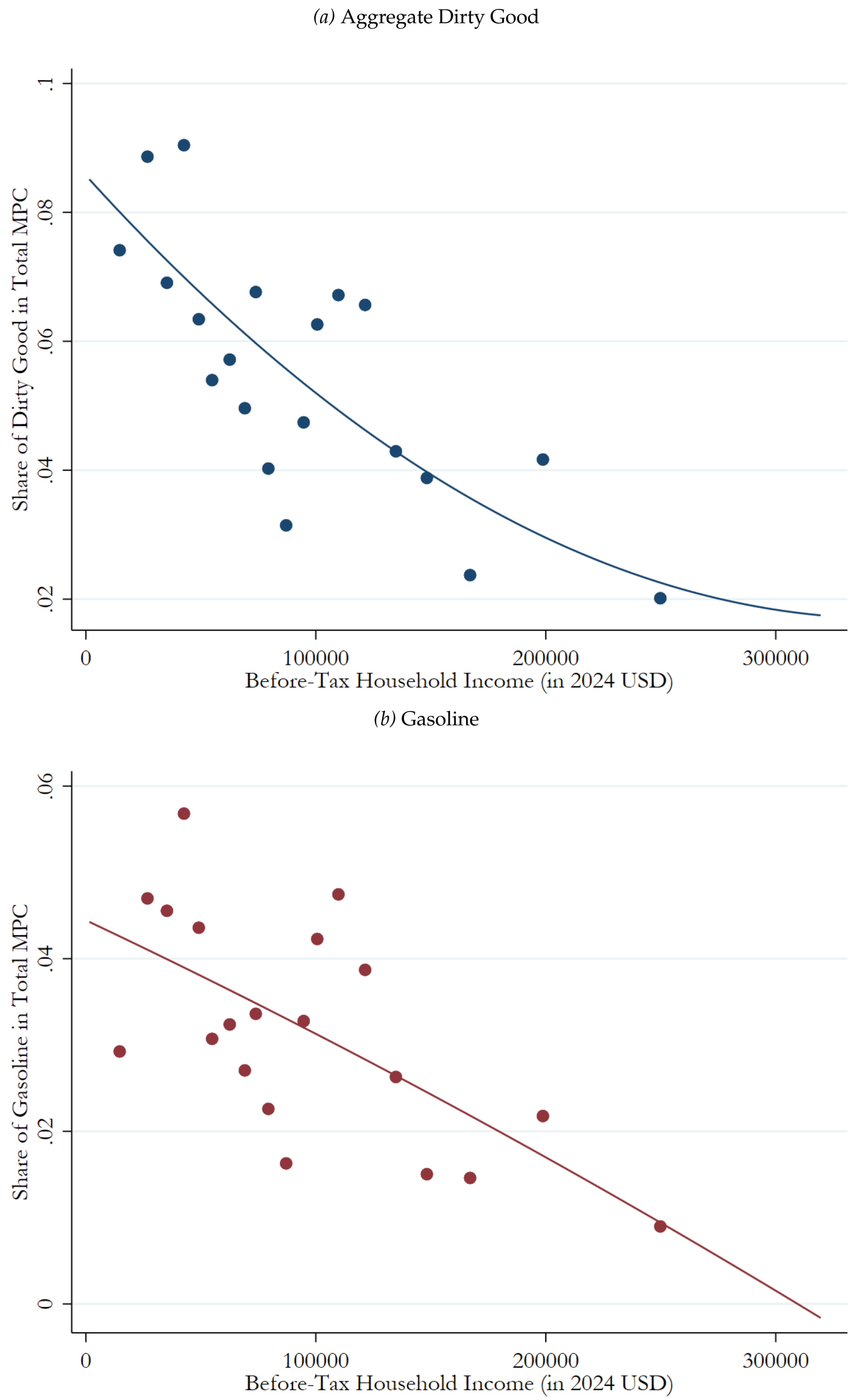


Figure 2. Causal income effects for carbon-intensive goods across the income distribution

Note: These binscatter plots show the estimated causal income effect $x'_{\text{inc}}(z)$ for carbon-intensive goods and the corresponding household taxable income, averaged across 20 bins of household taxable income (selected with *binsreg* (Cattaneo et al., 2024)), together with a global polynomial of degree 2, using data from a custom representative survey of the US population ($N = 1,448$). The "dirty good" expenditure category includes gasoline, electricity, natural gas, and heating fuels.

Imposing weak separability between income and consumption choices let us simplify this.[15] The causal effect of z on x then only occurs via the change in disposable income I, i.e. $\partial x/\partial z = \partial I/\partial z \times \partial x/\partial I|_{z=z(w)} = (1 - T'_z) \times \partial x/\partial I|_{z=z(w)}$, and we can rewrite the income effect on consumption of the dirty good holding taxable income fixed as:

$$\eta_{x|z}(z) = \frac{1}{1 - T'_z(z)} \frac{\partial x\left(\hat{w}\left(z\right); z\right)}{\partial z}. \tag{54}$$

This means the total impact of a pure income shock on consumption of x can be written as

$$\frac{dx\left(\hat{w}\left(z\right); z\right)}{dI} = \left(\frac{1}{1 - T'_z(z)} + \eta_z(z)\right) \frac{\partial x\left(\hat{w}\left(z\right); z\right)}{\partial z}. \tag{55}$$

If we further assume that the income effect on taxable income is negligible ($\eta_z \approx 0$), then

$$x'_{\text{inc}}(z) \equiv \frac{\partial x\left(\hat{w}\left(z\right); z\right)}{\partial z} = \left(1 - T'_z(z)\right) \frac{dx\left(\hat{w}\left(z\right); z\right)}{dI}. \tag{56}$$

4.1.3 Abstracting Away From Savings

Our model is static, and thus abstracts away from savings. However, households do save a fraction of their disposable incomes. We therefore make two adjustments. First, when constructing $\hat{x}'(z)$ with the CEX data, we use *shares* (as in Figure 1) rather than *levels* at each income percentile, and multiply these shares by the level of after-tax income. This results in a "savings-adjusted" level of consumption of the dirty good x.[16] Second, we use *shares* of total marginal propensity to consume (as depicted in Figure 2), i.e. the additional amount spent on carbon-intensive goods as a fraction of the total additional amount spent, rather than the dollar amount spent on carbon-intensive goods divided by $1,000.

4.1.4 Combining Cross-Sectional and Causal Variation To Estimate Taste Heterogeneity

Our next step is to smooth the data over the income distribution, project it to a fine income grid for numerical simulations, and calculate $x'_{\text{het}}(z)$ as the difference between $\hat{x}'(z)$ and $x'_{\text{inc}}(z)$. Starting from the discrete CEX percentile distribution for income z and dirty-good consumption x, we construct a "continuous" cross-sectional profile by fitting a smoothing spline to $\log x$ as a function of $\log z$ and then projecting the fitted values back to levels on a log-spaced income grid. We use the same income grid to evaluate the survey-based fitted regression equation for $x'_{\text{inc}}(z)$, mirroring the relationship plotted in Figure 2. Our

[15] There is limited evidence that gasoline consumption is complementary with leisure (West and Williams III, 2007), although this finding is not universal (Beznoska, 2014). If this is true for overall carbon intensity, our estimates understate the optimal tax on carbon-intensive goods. See Kaplow (2012) for further discussion.

[16] This effectively assumes that consumption shares remain constant over time, which is true with a utility function of the form: $U_i = u^i\left(f_i(c_1, x_1) + \beta f_i(c_2, x_2), z\right)$ where $f_i(c_t, x_t)$ is homothetic within each period $t \in \{1, 2\}$ and identical across periods. Note that in order to have a direct dependence of z on within-period consumption preferences, we need to allow the homothetic preference parameters to depend on z.

preferred specification uses a Poisson regression with log link and a quadratic in taxable income.[17] Finally, we differentiate the fitted cross-sectional profile $\hat{x}(z)$ to obtain $\hat{x}'(z)$ and subtract the predicted causal income-effect schedule $x'_{\text{inc}}(z)$ to obtain $x'_{\text{het}}(z)$.

The results are depicted in Figure 3. For brevity, we restrict the remainder of our empirical analysis to the aggregate dirty good. However, we present results for each component in Appendix A, which are very similar in their implications.

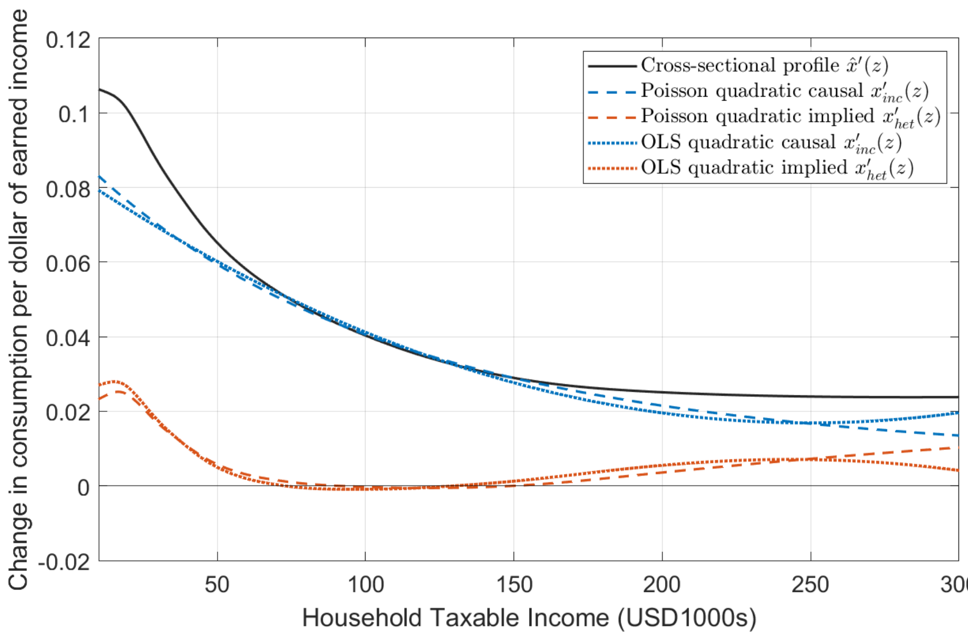


Figure 3. Decomposition of the cross-sectional profile $\hat{x}'(z)$ - Aggregate Dirty Good

Note: This figure shows how we decompose the cross-sectional relationship between income and carbon-intensive good consumption. The black line is a smoothed estimate of the cross-sectional slope, $\hat{x}'(z)$ at each level of income. The dashed blue lines show estimates of the causal effect of income on expenditure on carbon-intensive goods. The orange dashed lines subtract the income effect from the cross-section to obtain the part of the relationship between x and z that comes from preference heterogeneity.

The black curve in Figure 3 depicts the cross-sectional profile $\hat{x}'(z)$. It is positive, decreasing with income, but quickly flattens out for household taxable incomes between \$100,000 and \$300,000. This illustrates the increasing and concave relationship between *levels* of consumption of carbon-intensive goods and taxable income. The dashed blue curves show our fitted estimate of $x'_{\text{inc}}(z)$, which is decreasing throughout the income distribution. The dashed orange curves show the implied (residual) $x'_{\text{het}}(z)$ which determines the sign of $\eta^{\text{Taste}}_{x,z}(z)$, a central input into the shape of our optimal and Pareto-efficient

[17] See Table A.3 for details. Figure 3 shows that the Poisson quadratic and OLS quadratic fit imply very similar schedules, but we prefer the Poisson specification because its log link keeps the projected mean of x'_{inc} positive. We also regularize the top tail of the $x'_{\text{inc}}(z)$ schedule (starting at \$250k) to have it converge to the cross-sectional profile at very high incomes (at \$450k and above). This amounts to assuming a zero preference heterogeneity component at the very top of the income distribution.

nonlinear tax schedules. Our estimates of $\hat{x}'(z)$ and $x'_{\text{inc}}(z)$ are close overall. This leads to estimates of $x'_{\text{het}}(z)$ that are close to zero, which implies that differences in taste contribute very little to variation in carbon intensity over the income distribution.

It is important to note that there was no reason *a priori* to expect our estimate for the causal effect of income to closely match the cross-sectional profile across the income distribution. The first is computed as a slope using expenditures (i.e., levels) *across* households from the large CEX surveys, while the second is an individual-level response (i.e., a change) from our custom hypothetical survey. Their similarity is thus striking.

For direct application in our quantitative analysis, Figure 4 translates this $x'_{\text{het}}(z)$ into the taste elasticity that appears in Equation 31, for our preferred Poisson specification. We see that $\eta^{\text{Taste}}_{x,z}(z)$ is positive at low income levels, declines toward zero, and becomes slightly negative for a range of middle-income households. It then rises again and becomes positive at higher income levels. This schedule, together with the compensated elasticities of taxable income and dirty good consumption, the marginal damage, and the marginal income tax schedule, determines the shape of the Pareto efficient nonlinear tax schedule on consumption of the carbon-intensive good (see equation 30).

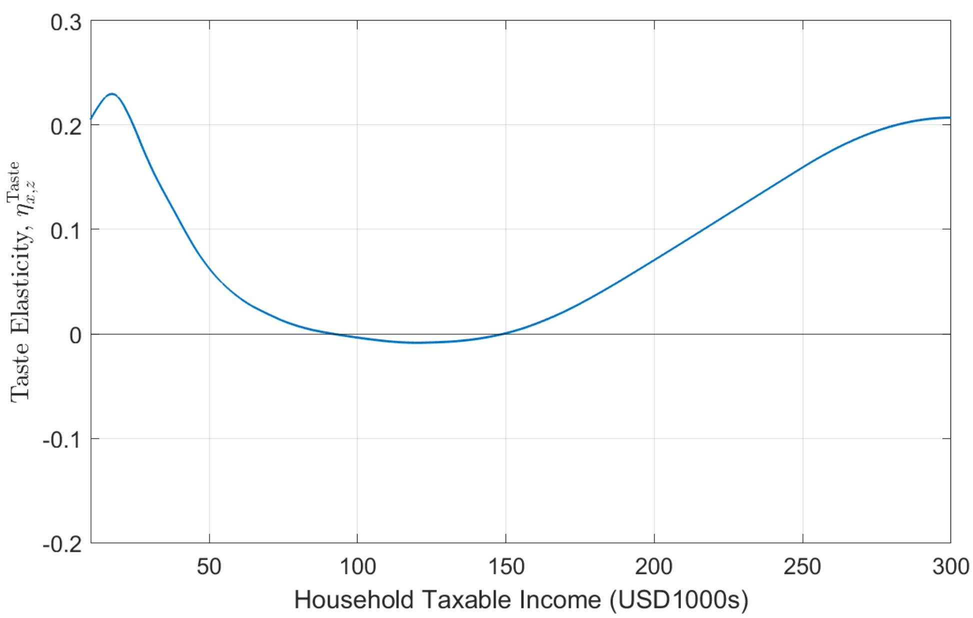


Figure 4. Elasticity of consumption of x with respect to income via taste heterogeneity only

Note: This figure plots the "taste elasticity", $\eta^{\text{Taste}}_{x,z}$. It is based on our estimates of $x'_{\text{het}}(z)$, which are shown by the dashed orange line in Figure 3 (the Poisson specification), but converted into an elasticity.

4.2 Estimating and Calibrating Other Quantities

Elasticity of consumption of carbon-intensive goods. We follow an empirical approach similar to that of Li et al. (2014). We construct a panel using the CEX, with variation over time and between states in gasoline excise taxes and prices. Then we study the consumption response of gasoline to tax changes from 1996–2021. We estimate the uncompensated elasticity to be 0.59 (SE: 0.30) in our preferred specification with month$\times$year fixed effects and household-level controls (see Appendix A.5 for details). We thus use three values for the compensated elasticity in our tax simulations for sensitivity analysis: 0.5 for our preferred *benchmark* scenario, 0.25 for the *low* elasticity of dirty good consumption scenario, and 0.75 for the *high* elasticity of dirty good consumption scenario.

Elasticity of taxable income. We calibrate the elasticity of taxable income $\varepsilon_z(z)$ for our *benchmark* scenario to be constant at 0.33, which is the preferred estimate (on the intensive margin) from the meta-analysis of Chetty (2012). We also consider a value of 0.7 for our *high* taxable income elasticity scenario.

Marginal Damage From Carbon. We follow Hahn et al. (2024) and assume a Social Cost of Carbon (SCC) of \$200 per ton. This is within the range of estimates provided by a recent report of the US Environmental Protection Agency (EPA, 2023). We assume that consumption of carbon-intensive goods generates an average of 2kg of CO_2 per dollar of final expenditure.[18] This results in a marginal damage of 40 cents per dollar of final expenditure, which implies a baseline Pigouvian tax of 40% in our model[19]. We note that our Pareto-efficient tax schedules are calibrated relative to this marginal damage estimate. A reader who prefers an alternative estimate of the SCC can simply shift the levels of all of our optimal tax schedules (e.g., Figure 6) to match their higher or lower estimate of the marginal damage from carbon-intensive consumption.

Marginal income tax rate schedule. We compute the average before- and after-tax household income for each income percentile using data from the 2018 to 2024 CEX quarterly waves. We convert these values to 2024 USD using the average annual Consumer Price Index (CPI). We then smooth the relationship between after-tax and before-tax income, fit a smoothing spline and compute its derivative to recover the marginal tax rates on taxable income. The blue curve in Figure A.4 shows the resulting marginal income tax rates across the income distribution used in our simulations.

4.3 Empirically Optimal Tax Rates

Figure 5 plots the optimal marginal tax rates on the carbon-intensive good implied by our Pareto efficiency conditions for the unidimensional case. Panels (a) and (c) present the

[18]The aggregate dirty good has average CO_2 emissions of 2.24kg per dollar of final expenditure, calculated as the expenditure-weighted average of its three components for which emissions data are available: gasoline, electricity, and natural gas (see Appendix Table A.2 for a breakdown by component).

[19]This approximately corresponds to the gasoline tax in France (TICPE) in 2024 (DILA, 2024); while in our data (see Appendix Section A.5 for data sources) the average gasoline tax across US states was 18.5% in 2021.

nonlinear case (equation 30). Panels (b) and (d) show the case where the tax is restricted to be linear (equation 34). Panels (a) and (b) assume an elasticity of taxable income $\varepsilon_z = 0.33$, which is most consistent with Chetty's (2012) meta-analysis. Panels (c) and (d) assume $\varepsilon_z = 0.7$. In each panel, we present estimates for a range of values for the compensated net-of-tax rate elasticity of spending on the carbon-intensive good.

4.3.1 Optimal Nonlinear Taxation of Carbon — Unidimensional Case

We focus first on the nonlinear case. Throughout most of the income distribution, the Pareto-efficient carbon tax is remarkably close to the marginal damage from these goods (the Pigouvian level). The main deviation from this is that the optimal carbon tax rises above the Pigouvian level for the highest-income households. There is also a much smaller increase compared to the Pigouvian benchmark for a range of low-income households. We note that the fact that the optimal tax is some times higher but never more than marginally lower than the Pigouvian level contrasts sharply with the naive logic that carbon taxes should be set lower simply because they are regressive.

This pattern is amplified when consumption of carbon-intensive goods is assumed to be very inelastic (i.e., $\varepsilon_{x|z}$ is very low), especially if the elasticity of taxable income is very high (e.g., $\varepsilon_z = 0.7$ as in panel c). That combination of parameters implies that income taxation is highly distortionary relative to adjusting the carbon tax for distributional reasons, making the latter a lower cost redistributive instrument. However, this is only true because the part of the covariance of income and carbon intensity that is explained by differences in preferences ($\eta_{x,z}^{\text{Taste}}(z)$) is positive at high levels of income.

4.3.2 Optimal Linear Taxation of Carbon — Unidimensional Case

Panels (b) and (d) show the results when we restrict to linear taxation. The Pareto-efficient linear tax is close to the Pigouvian level for all parameter values. In our benchmark scenario with $\varepsilon_z = 0.33$, and $\varepsilon_{x|z} = 0.5$, the Pareto-efficient linear tax is just 2.9 percent higher than marginal damage. In this sense, distributional concerns have very little bearing at all on the optimal carbon tax if it is required to be linear.

This stark result comes from the fact that the Pareto-efficient linear tax is a weighted average of the quantities determining the marginal tax rates at each income level in the nonlinear case (see equation 34). Intuitively, departures from the Pigouvian level of the carbon tax will depend on a weighted average tagging benefit of carbon. As Figure 4 shows, $\eta_{x,z}^{\text{Taste}}(z)$ is close to zero for much of the income distribution, especially in the middle range where most taxpayers are located. Integrating over the distribution, the Pareto-efficient linear tax is therefore above, but very close to, the Pigouvian level.

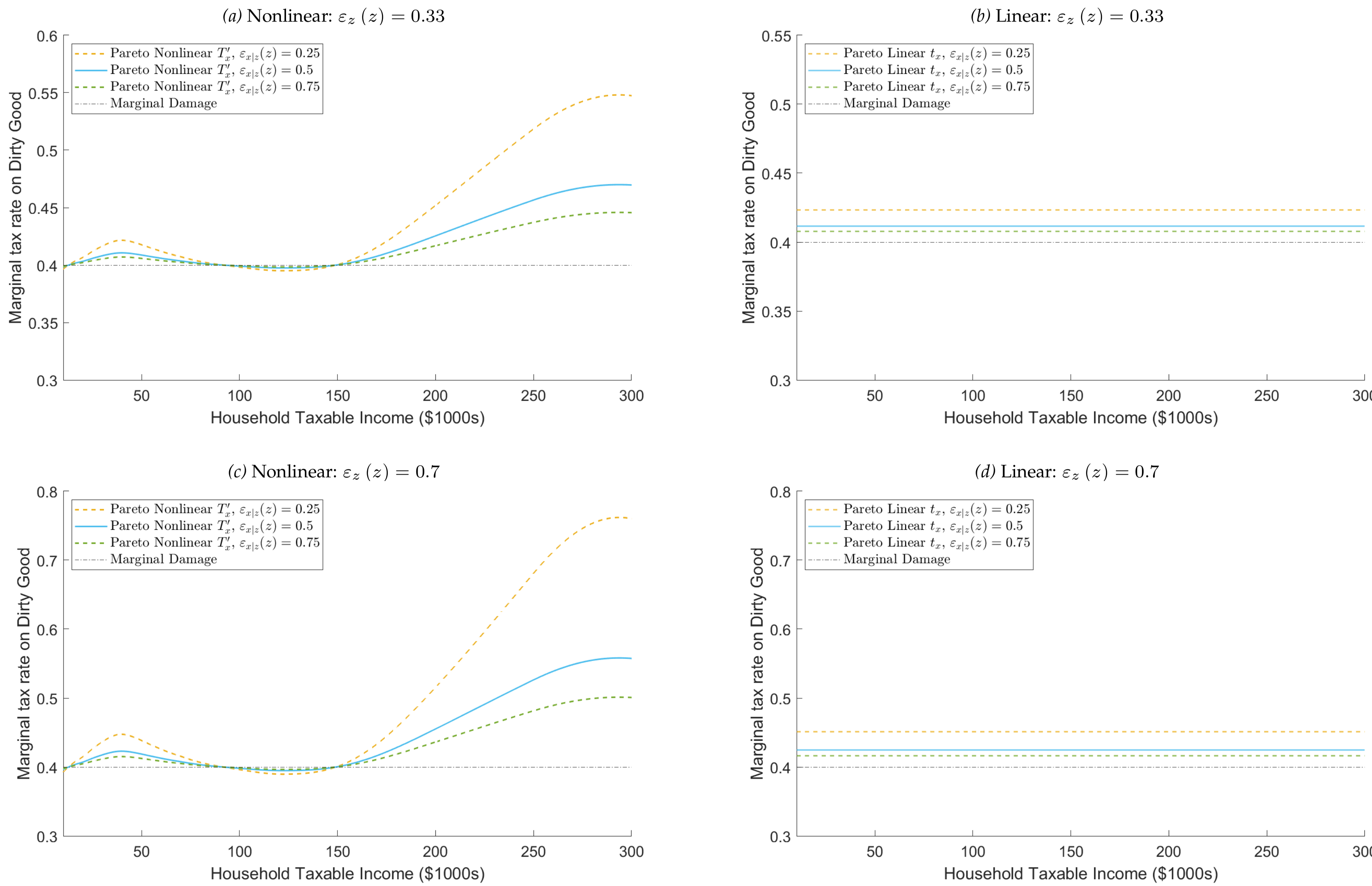


Figure 5. Dirty Good Tax Schedule: Marginal Rates implied by Pareto Efficient formulas

Note: This figure plots the Pareto efficient carbon tax for different values of the elasticities of taxable income and consumption of the carbon-intensive good. Three cases are shown in each panel, along with the benchmark of the assumed value of marginal damage. Panels (a) and (c) show nonlinear marginal tax schedules, while panels (b) and (d) restrict the carbon tax to be linear.

4.3.3 Optimal Linear Taxation of Carbon — Multidimensional Case

To quantify the optimal carbon tax with multidimensional heterogeneity, we need the conditional variance of the income effect. To estimate this, we exploit our individual-level survey. Specifically, we use the squared residuals from the estimation of the conditional mean $x'_{\text{inc}}(z)$ with our Poisson-quadratic specification.[20] This lets us estimate $\text{Var}(x'_{\text{inc}} \mid z)$ by fitting the squared residuals with a quadratic generalized linear regression in taxable income (see Appendix A.4). We project this fitted conditional variance schedule onto the log-spaced income grid we used for the unidimensional numerical simulations.

Table A.4 shows the regression results, while Figure A.3 plots the fitted schedule. The conditional variance of the causal income effect is largest for households at the bottom of the income distribution, and declines moderately over the income range shown, implying that multidimensional heterogeneity in the causal income response is most significant at lower incomes and smaller at higher income levels.[21]

Aside from $\text{Var}(x'_{\text{inc}} \mid z)$, we re-use most of the empirical quantities from the unidimensional case, with the difference that point estimates used for calibration of the unidimensional case are interpreted as conditional averages. For example, $\eta^{\text{Taste}}_{x,z}(z) = \mathbb{E}[\eta^{\text{Taste}}_{x,z}(z,\theta) \mid z]$ and $\bar{x}(z) = \mathbb{E}[x(z,\theta) \mid z]$. We continue to assume constant values for the elasticities of taxable income and consumption of the carbon-intensive good.

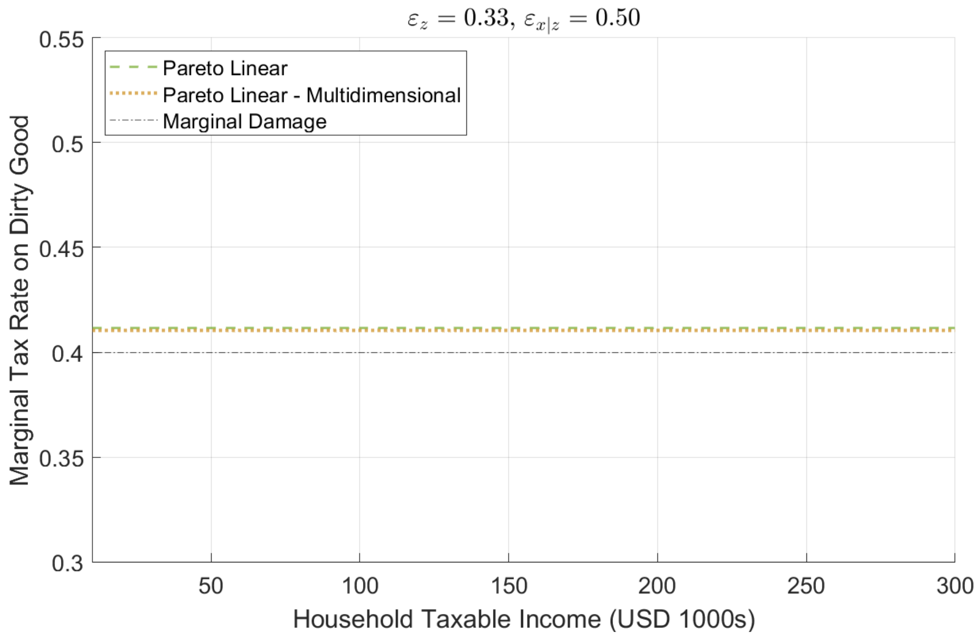
Figure 6 plots the optimal tax rate for the multidimensional case (orange dotted line). For comparison, we also include the linear tax rate implied by the Pareto efficient linear tax formula in the unidimensional case (green dashed line). Panel (a) assumes an elasticity of taxable income $\varepsilon_z = 0.33$, and an elasticity of consumption of the carbon-intensive good $\varepsilon_{x|z} = 0.5$, while panel (b) assumes $\varepsilon_z = 0.7$, and $\varepsilon_{x|z} = 0.25$. As expected based on our theoretical results, within-income variation of the causal effect of income leads to attenuation: The optimal tax is pushed closer to the Pigouvian benchmark.

The degree of attenuation is larger in the lower panel, where the deviation from Pigouvian taxation was larger without multidimensional heterogeneity. This is because the amount of attenuation due to $\text{Var}(x'_{\text{inc}} \mid z)$ is proportional to the ratio of elasticities $\varepsilon_z/\varepsilon_{x|z}$ (see equation 50). Recall that this same ratio scales the intensity of the tagging benefit in the first place, which implies that the degree of attenuation from within-income heterogeneity in $\partial x/\partial z$ is greatest when the tagging benefit is largest. As a result, there is only minimal attenuation in Figure 6(a) where the optimal linear rate in unidimensional

[20] Said differently, we exploit the identity $e_i^2 = \left(x'_{\text{inc},i} - \widehat{\mathbb{E}}[x'_{\text{inc}} \mid z_i]\right)^2$.

[21] Computing the raw variance of the causal effect at deciles of the income distribution yields similar values to the fitted curve. We note that both approaches may yield upwardly biased estimates of $\text{Var}(x'_{\text{inc}} \mid z)$ if there is significant measurement error in x'_{inc}, which is plausible.

(a) Baseline calibration



(b) Higher ε_z, lower $\varepsilon_{x|z}$

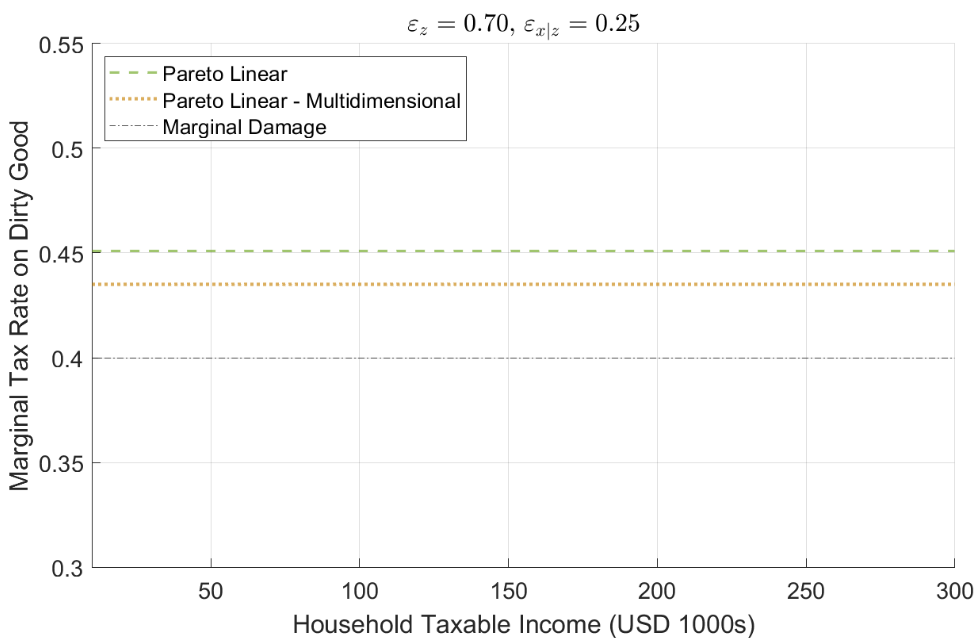


Figure 6. Dirty Good Marginal Linear Tax Rate: Unidimensional vs. Multidimensional Heterogeneity

Note: This figure plots the Pareto efficient (linear) carbon tax in the unidimensional case vs. the optimal linear tax rate when accounting for multidimensional heterogeneity for different values of the elasticities of taxable income and consumption of the carbon-intensive good.

case was almost exactly at the Pigouvian level already. There is more attenuation in panel (b), where the higher ratio of elasticities led to a larger deviation.

5 Conclusion

The use of taxes or subsidies to internalize externalities is widely accepted by economists as an important role of government. When it comes specifically to climate change, economists have forcefully advocated for a carbon tax on these grounds (Akerlof et al., 2019). But this idea has gained limited traction politically.

We formally study one specific critique of carbon taxes, and of corrective taxes more generally: They can be regressive. Our theoretical results show that the fact that poorer individuals are more carbon-intensive in their consumption does not necessarily change the optimal carbon tax. What matters is whether this regressivity reflects income effects or preference heterogeneity. Differences in taste for carbon-intensive goods at different income levels can make the carbon tax a lower-cost redistributive instrument relative to the income tax. We show how to quantify preference heterogeneity empirically, and take our sufficient statistics tax formulas to the data.

The implications of our empirical results for the optimal carbon tax depend on whether nonlinear carbon taxation is feasible. If only linear carbon taxation is possible, then the optimal carbon tax is remarkably close to the classic Pigouvian prescription of setting the tax equal to the marginal damage from emitted carbon. This conclusion for the optimal linear tax is robust to a very wide range of different parameter values.

The conclusion is more nuanced if nonlinear carbon taxation is possible. Our benchmark scenario implies that optimal marginal tax rates are close to the Pigouvian level for most households, but above that level for higher-income households. This pattern is amplified if the elasticity of taxable income is higher, or the behavioral response to carbon taxation is smaller. In such cases, adjusting the carbon tax becomes a more attractive tool to redistribute income, relative to income taxation.

Our conceptual and empirical frameworks could be applied to many settings that involve both distributional concerns and externalities. This could include other environmental externalities, public health problems, or even criminal behavior. Our results suggest that preference-driven differences in carbon intensity have a limited impact overall. Our estimate of the optimal carbon tax is therefore close to the Pigouvian benchmark. However, this result may differ for other externalities. In the future, it will also be important to integrate our detailed work on the consumption side of the economy with richer analysis of the production side along the lines of Bierbrauer (2024) and others.

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Appendix - Table of Contents

A Empirical Appendix

A.1 Average Expenditure Shares and CO_2 Emissions across Categories

Table A.1. Average Expenditure Shares across Categories

	Avg.
Food and Beverages	0.186
Housing (excluding utilities)	0.268
Electricity, natural gas, and home heating fuels	0.045
Water, sewage, and other public services	0.013
Healthcare	0.092
Gasoline and other motor fuels	0.041
Flights / Airfare expenditures	0.005
Other Expenditures	0.351
Observations	112,885

Note: This table shows shares allocated to each consumption category among consumers in the Consumer Expenditure Survey tax-schedule working sample: 2018–2023 plus 2024Q1.

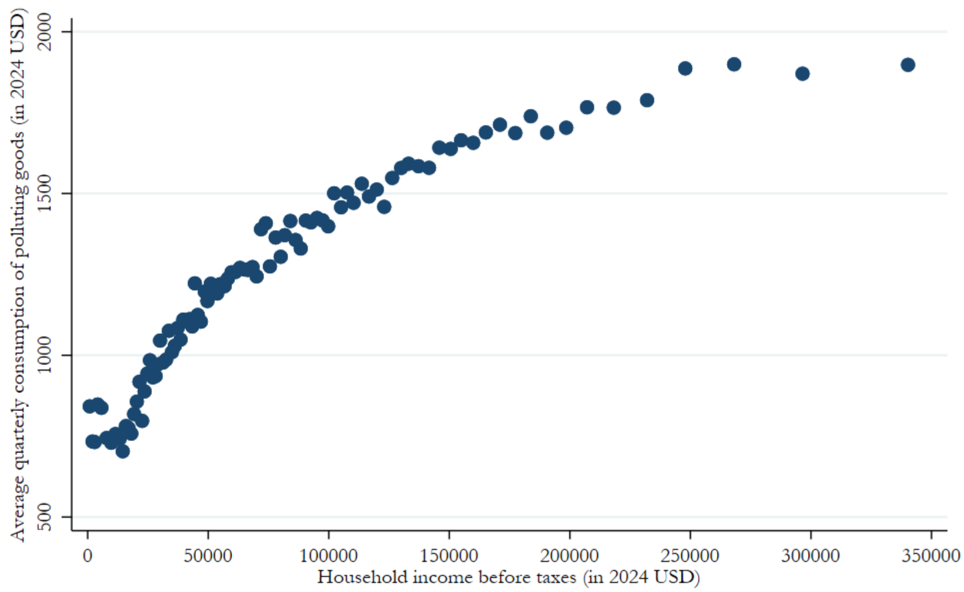
Table A.2. Average CO_2 Emissions per Dollar of Final Energy Expenditure in the United States

	Unit	**Price per Unit (2023 $)**	**CO_2 Emissions per Unit (kg)**	**CO_2 Emissions (kg per $)**
Gasoline	Gallon (3.785 litres)	3.635	8.778	2.415
Electricity	kWh delivered	0.1600	0.363	2.270
Natural gas	1,000 cubic feet (Mcf)	15.40	54.440	3.535
Air travel	Passenger-mile	0.1566	0.182	1.161

Note: Prices are nominal 2023 U.S. average final consumer prices from the U.S. Energy Information Administration (EIA). Gasoline uses the EIA U.S. annual retail price for all grades. Electricity and natural gas use EIA residential prices, consistent with household final expenditure categories. Emissions are direct CO_2 emissions only and exclude upstream, refining, transmission, and other life-cycle emissions. Electricity emissions use EPA eGRID2023's U.S. average CO_2 output rate, adjusted for grid gross losses. Gasoline and natural gas emissions use default combustion factors. Air travel emissions are calculated using jet fuel emission factors (EIA/EPA) and passenger-mile fuel economy from the U.S. Transportation Energy Data Book. The price per passenger-mile uses FAA passenger yield data.

A.2 Carbon-Intensive Consumption in *Levels*

(a) Aggregate Dirty Good



(b) Gasoline

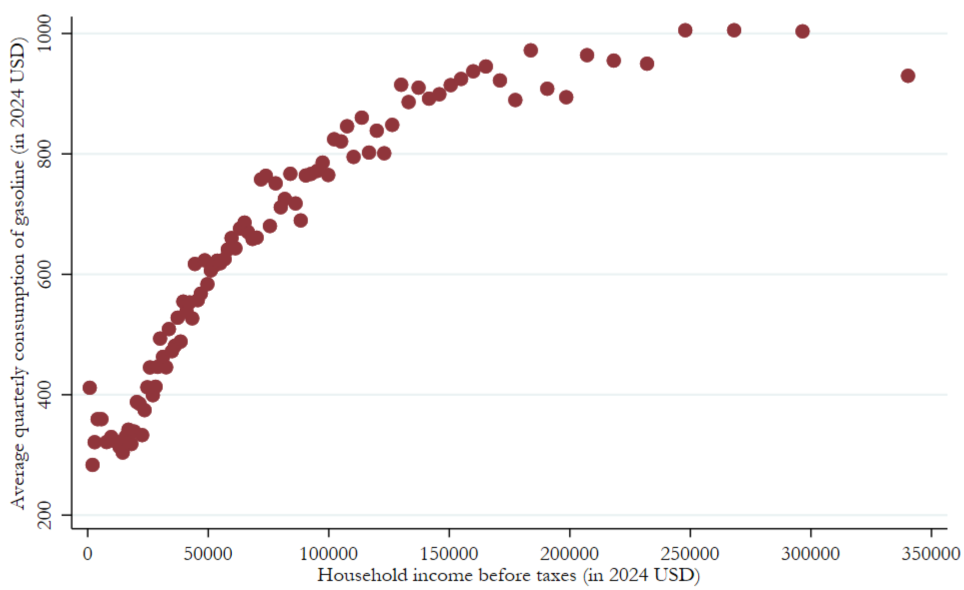
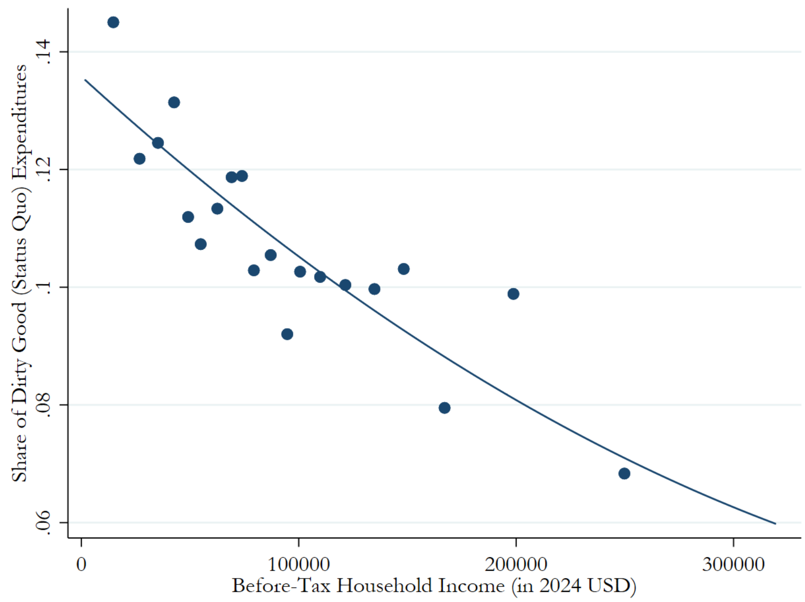


Figure A.1. Quarterly consumption expenditure of carbon-intensive goods across the income distribution

Note: These binscatter plots show (i) the average quarterly expenditures on polluting goods, and (ii) the corresponding average household taxable income, for each income percentile using data from the 2018–2024 Consumer Expenditure Survey quarterly waves. The "dirty good" expenditure category includes gasoline, electricity, natural gas, and heating fuels. Household consumption expenditures and taxable incomes were converted to 2024 USD using the average annual Consumer Price Index in the US. The figure is truncated at $350k for readability. This excludes the top 3% of earners.

A.3 Share of Carbon-Intensive Goods in Status Quo Expenditures

(a) Aggregate Dirty Good



(b) Gasoline

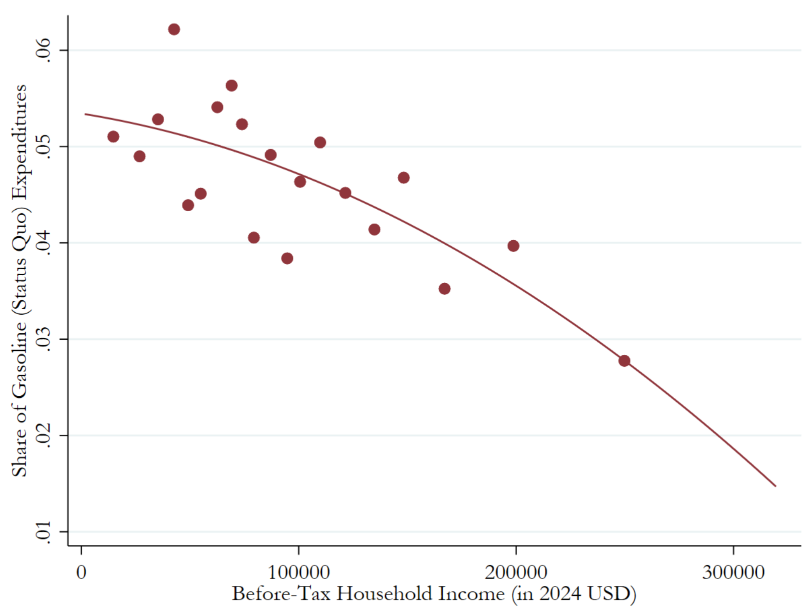


Figure A.2. Share of carbon-intensive goods in total (monthly) expenditures in custom survey

Note: These binscatter plots show (i) the average share of expenditure on polluting goods as a fraction of total expenditures, and (ii) the corresponding average household taxable income, averaged across 20 bins of household taxable income (selected with the command binsreg (Cattaneo et al., 2024)), and (iii) a global polynomial of degree 2 estimated to fit the relationship between these two variables, using data from a custom representative survey of the US population ($N = 1,448$). The "dirty good" expenditure category includes gasoline, electricity, natural gas, and heating fuels.

A.4 Estimating Income-Conditional Moments of the Causal Consumption Response $\mathbb{E}[x'_{\text{inc}}|z]$ and Var$[x'_{\text{inc}}|z]$

Table A.3. Coefficient Estimates – Conditional Mean of the Causal Income Effect $x'_{\text{inc}}(z)$

	Dirty goods		Elec. and heating fuels		Gasoline	
	OLS	Poisson	OLS	Poisson	OLS	Poisson
Income (thousands USD)	-.000542***	-.0089***	-.000325***	-.0141***	-.000217**	-.00428
	(.000125)	(.00276)	(.0000735)	(.00328)	(.0000865)	(.00362)
Income squared	1.09e-06***	8.51e-06	8.10e-07***	.0000305**	2.77e-07	-.000012
	(4.07e-07)	(.0000116)	(2.41e-07)	(.0000126)	(2.76e-07)	(.0000155)
Constant	.0846***	-2.4***	.0407***	-3.04***	.0438***	-3.14***
	(.00796)	(.133)	(.00472)	(.165)	(.00556)	(.177)
Mean squared error	0.00812	0.00812	0.00239	0.00239	0.00402	0.00402
Corr. with fitted values	0.182	0.184	0.165	0.173	0.138	0.138
Mean Dep. Var.	0.047	0.047	0.020	0.020	0.027	0.027
SD Dep. Var.	0.092	0.092	0.050	0.050	0.064	0.064
Observations	1,448	1,448	1,448	1,448	1,448	1,448

Note: This table reports coefficient estimates from the regression of $x'_{\text{inc},i} = s^{\text{MPC}}_{g,i}(1 - T'_{z,i})$ on before-tax income z and before-tax income squared z^2, where s^{MPC}_g is the share of marginal propensity to consume allocated to good g and T'_z is the status-quo marginal tax rate matched by before-tax household income. Columns are estimated separately for each good using our custom survey sample. Income is measured in thousands of 2024 dollars. *OLS* columns correspond to a linear regression with a quadratic in income; *Poisson* columns correspond to a log-link quadratic specification. Robust standard errors are in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.4. Coefficient Estimates – Conditional Variance of the Causal Income Effect $\mathrm{Var}(x'_{\mathrm{inc}} \mid z)$

	Dirty goods	Elec. and heating fuels	Gasoline
Income (thousands USD)	-.00555	-.0126*	-.00292
	(.00424)	(.00671)	(.00621)
Income squared	-8.82e-06	.0000219	-.0000268
	(.0000164)	(.0000271)	(.0000236)
Constant	-4.26***	-5.21***	-5.03***
	(.229)	(.345)	(.345)
Mean squared error	0.00050	0.00007	0.00032
Corr. with fitted values	0.131	0.123	0.086
Avg. predicted var. / avg. residual sq.	1.000	0.995	1.001
Observations	1,448	1,448	1,448

Note: This table reports Gamma GLM coefficient estimates with a log link for the squared residuals from the baseline Poisson-quadratic conditional-mean model of x'_{inc}, shown in Table A.3. The fitted values provide the paper's estimate of $\mathrm{Var}(x'_{\mathrm{inc}} \mid z)$ as a function of before-tax household income z. Income is measured in thousands of 2024 dollars. The row "Avg. predicted var. / avg. residual sq" reports the ratio of the average predicted variance (fitted values from the regression) to the average squared residual from the baseline Quadratic Poisson model (see Table A.3). Standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

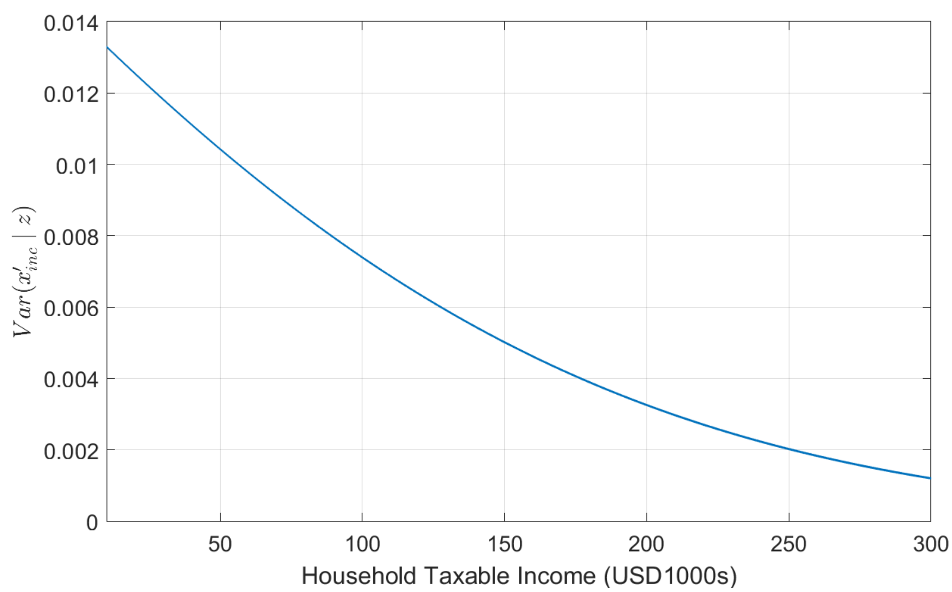


Figure A.3. Conditional variance of $x'_{\mathrm{inc}}(z)$ across the income distribution

Note: This figure shows the fitted conditional variance of the causal effect of income on dirty-good consumption, $\mathrm{Var}(x'_{\mathrm{inc}} \mid z)$, estimated from our custom survey ($N = 1,448$). The fitted curve comes from a log-link Gamma GLM for squared residuals from the preferred Poisson quadratic mean specification and is evaluated on the same income grid used in the tax simulations.

A.5 Estimating the Uncompensated Elasticity of Gasoline Consumption

Following Li et al. (2014), we estimate a log-log two-way fixed effects regression to recover the elasticity of gasoline consumption with respect to the net-of-tax rate. We use granular data at the household level rather than aggregated data at the state level and estimate the following regression equation:

$$\log(q_{ist}) = \alpha \log(p_{st}) + \beta \log(1 + \frac{\tau_{st}}{p_{st}}) + \Theta X_{ist} + \delta_s + \phi_t + e_{ist} \tag{57}$$

where q_{ist} is the household-level consumption of gasoline (in quantities), p_{st} is the tax-exclusive price of gasoline, τ_{st} is the gasoline excise tax (in dollars); X_{ist} is a vector of household-level controls; and δ_s and ϕ_t are state, and time (yearly or monthly) fixed effects.

The coefficient $-\beta$ in the first row can be interpreted as the uncompensated elasticity of gasoline consumption with respect to the net-of-tax rate.

Table A.5. Elasticity of gasoline consumption with respect to tax rate: 1996-2021

	(1)	(2)	(3)	(4)
$\log(1 + \text{tax ratio})$	-0.379	-0.382	-0.585**	-0.589**
	(0.322)	(0.322)	(0.297)	(0.297)
$\log(\text{gas price})$	-0.453***	-0.454***	-0.414***	-0.415***
	(0.145)	(0.145)	(0.136)	(0.136)
State FE	Y	Y	Y	Y
Year FE	Y		Y	
Month × Year FE		Y		Y
Household Level Controls	N	N	Y	Y
R^2	0.089	0.090	0.201	0.202
Observations	596,822	596,822	542,574	542,574
Clusters (State × Year)	1,035	1,035	991	991

Note: This table shows the coefficient estimates from the regression of equation (57). All regressions control for family size and use annual weights to make the sample representative of the United States each calendar year (see section 6.3.1 of `https://www.bls.gov/cex/pumd-getting-started-guide.htm#section3`). Additional household-level control variables include age (of the reference person), urban vs. rural status, taxable income, education level, and family type. Quarterly data from 1996–2021 on household gasoline consumption are from the interview module of the Consumer Expenditure Surveys (CEX) Public use microdata files. We can use Month × Year fixed effects because of the rotating nature of the survey. Gasoline tax data are obtained from the U.S. Department of Transportation, Federal Highway Administration, and gasoline prices are from the U.S. Energy Information Administration (State Energy Data System, SEDS) for the same period. Standard errors are clustered at the State × Year level. *** p<0.01, ** p<0.05, * p<0.1.

A.6 Status Quo Income Marginal Tax Rates

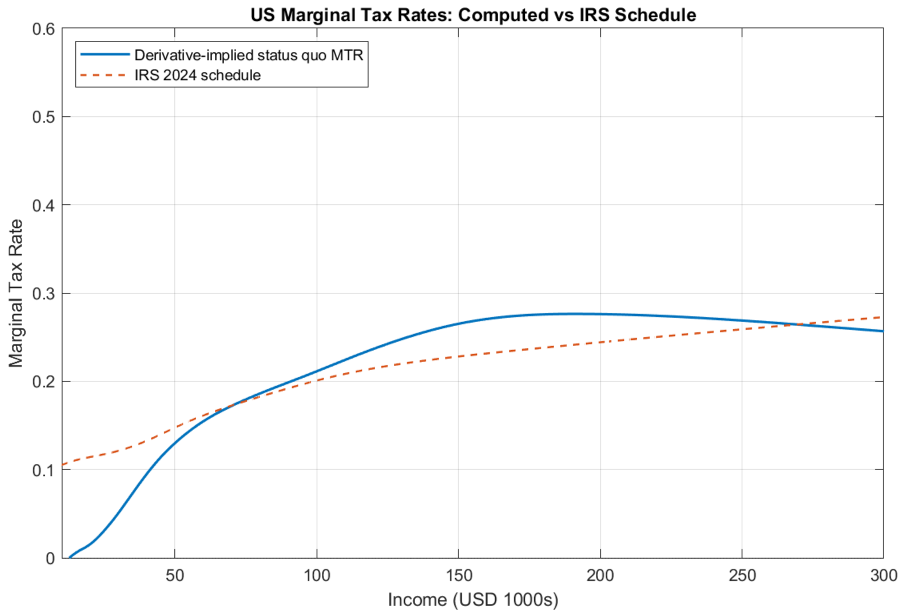


Figure A.4. Status quo income marginal tax rates across the income distribution

Note: This figure shows the status quo income marginal tax rates across the household income distribution. The blue curve shows estimated marginal tax rates used in our analysis, and computed from CEX data (2018–2024, converted to 2024 USD) by fitting a smoothing spline to the relationship between after-tax and before-tax income binned by income percentiles, and computing 1 minus its derivative. The dashed orange line shows the 2024 IRS statutory tax rate schedule (adjusted to account for the share of married couples) for illustrative comparison.

A.7 Pareto Efficient Marginal Tax Schedules - Gasoline Only

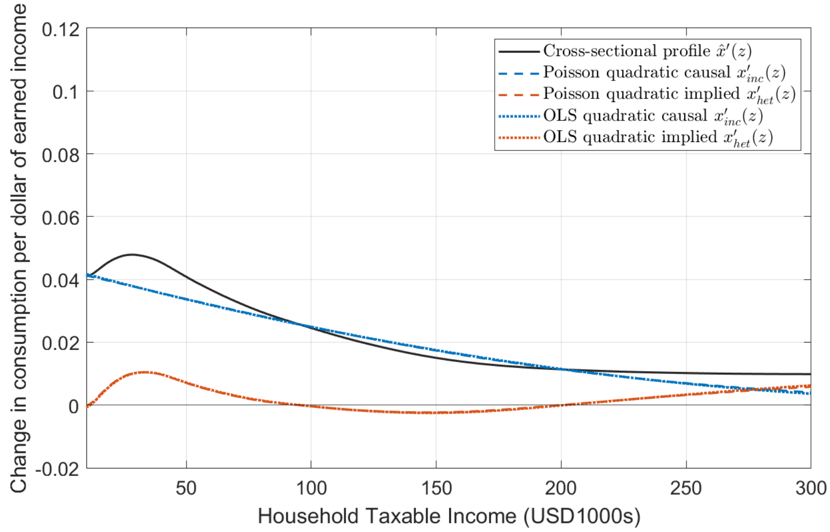


Figure A.5. Gasoline: Decomposition of the cross-sectional profile $\hat{x}'(z)$

Note: This figure shows how we decompose the cross-sectional relationship between income and **gasoline consumption**. The solid black line is a smoothed estimate of the cross-sectional slope, $\hat{x}'(z)$, at each level of income (from the 2018–2024 CEX surveys). The dashed blue lines show estimates of the causal effect of income on expenditure on gasoline (from our custom hypothetical survey). The orange dashed lines subtract the income effect from the cross-section to obtain the part of the relationship between x and z that comes from preference heterogeneity.

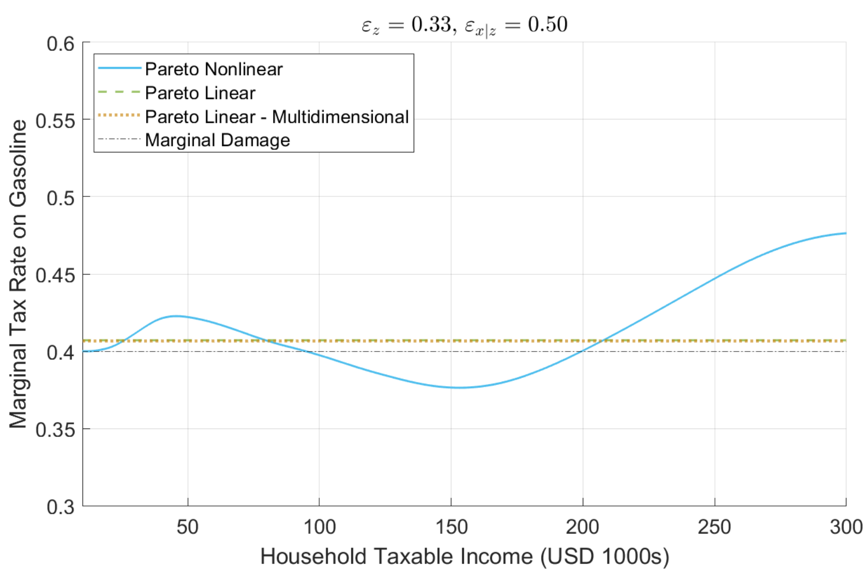


Figure A.6. Gasoline Marginal Tax Rates: Unidimensional vs. Multidimensional Heterogeneity

Note: This figure plots the Pareto efficient carbon tax for **gasoline only** in the unidimensional case (Linear and Nonlinear) vs. the optimal linear tax rate when accounting for multidimensional heterogeneity when the elasticity of taxable income is $\varepsilon_z = 0.33$, and the elasticity of consumption of the carbon-intensive good $\varepsilon_{x|z} = 0.5$.

A.8 Pareto Efficient Marginal Tax Schedules - Electricity, Natural Gas, and Heating Fuels Only

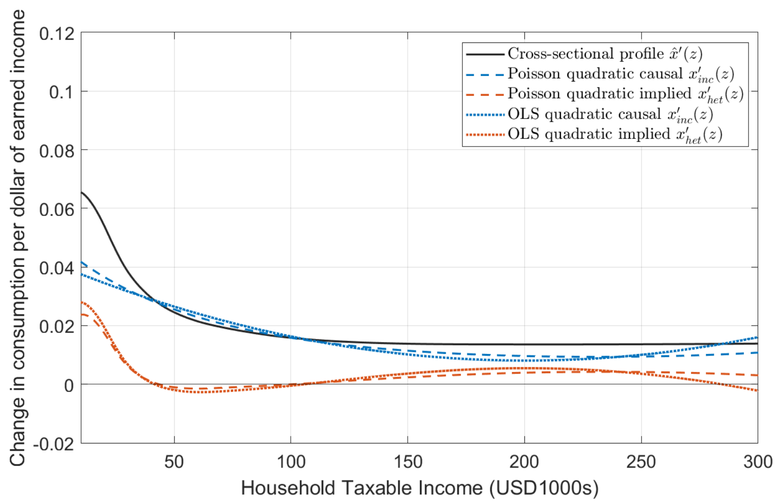


Figure A.7. Electricity, Natural Gas, and Heating Fuels: Decomposition of the cross-sectional profile $\hat{x}'(z)$

Note: This figure shows how we decompose the cross-sectional relationship between income and consumption of **electricity, natural gas, and heating fuels**. The solid black line is a smoothed estimate of the cross-sectional slope, $\hat{x}'(z)$, at each level of income (from the 2018–2024 CEX surveys). The dashed blue lines show estimates of the causal effect of income on expenditure on the same goods (from our custom hypothetical survey). The orange dashed lines subtract the income effect from the cross-section to obtain the part of the relationship between x and z that comes from preference heterogeneity.

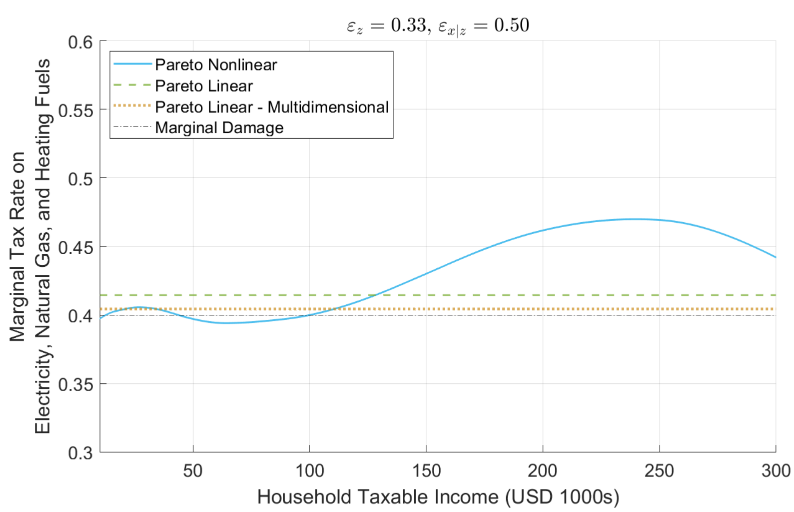


Figure A.8. Electricity, Natural Gas, and Heating Fuels Marginal Tax Rates: Unidimensional vs. Multidimensional Heterogeneity

Note: This figure plots the Pareto efficient carbon tax for **electricity, natural gas, and heating fuels consumption only** in the unidimensional case (Linear and Nonlinear) vs. the optimal linear tax rate when accounting for multidimensional heterogeneity when the elasticity of taxable income is $\varepsilon_z = 0.33$, and the elasticity of consumption of the carbon-intensive good $\varepsilon_{x|z} = 0.5$.

A.9 Online Survey Demographics Breakdown & Sample Definition

Ethnicity	Freq.	%	(C)
White	1097	75.8%	73.9%
Black	199	13.7%	14.3%
Asian	87	6.0%	7.5%
Mixed	43	3.0%	2.5%
Other	22	1.5%	1.7%

Age	Freq.	%	(C)
25-29	263	18.2%	16.9%
30-34	276	19.1%	18.0%
35-39	294	20.3%	17.3%
40-44	196	13.5%	16.8%
45-49	223	15.4%	15.2%
50-54	196	13.5%	15.9%

Sex	Freq.	%	(C)
Female	748	51.7%	49.7%
Male	700	48.3%	50.3%

Table A.6. Demographics Comparison: Online Survey vs 2023 Census Population Estimates

Note: This table shows the breakdown for our custom online survey sample by ethnicity, age, and sex. The rightmost (C) column in each subtable corresponds to the actual demographic distribution of the US population aged 25-54 drawn from 2023 Population Estimates (as of July 1) of the United States Census Bureau constructed with data from the 2020 Census and measures of population change. (See `https://www.census.gov/data/tables/time-series/demo/popest/2020s-national-detail.html`.)

Table A.7. Online Survey Response Quality: Sample Exclusions

	N	%
Total # of Respondents	2,214	100.00
Suspected Bots	18	0.81
Speedsters	421	19.02
Failed Attention Check	44	1.99
Inconsistent or Outlier Income Response	50	2.26
Zero Expenditure, Zero or Inconsistent MPC	233	10.52
Final Sample	1,448	65.40

Note: This table reports the sequential exclusion of Prolific respondents from the combined Waves 1 and 2 of our online survey. Exclusions are applied in the order shown, as per our pre-registration, with each category excluding observations from the remaining sample after previous exclusions. Percentages are calculated relative to total observations.

B Optimal Levels of the Income and Externality Taxes

An approach very similar to the one used to derive the Pareto-efficiency conditions in the main text can be used to characterize the optimal levels of the externality and income taxes. The fact that a reform to each tax schedule is not by itself distribution-neutral means that they involve trade-offs between individuals, and therefore marginal social welfare weights. We follow Diamond (1975) and Ferey et al. (2024) in defining augmented welfare weights:

$$g(w) = \gamma(w)\frac{u_c(c(w), x(w), z(w) \mid w)}{\lambda} + \left[\mathcal{T}_x'\left(x\left(w\right)\right) - \frac{\mathcal{D}'\left(\bar{x}\right)}{\lambda}\right] \left.\frac{\partial \hat{x}(w)}{\partial I}\right|_{z=z(w)}$$
$$+ \left\{\mathcal{T}_z'(z(w)) + \left[\mathcal{T}_x'(x\left(w\right)) - \frac{\mathcal{D}'\left(\bar{x}\right)}{\lambda}\right] x_{\text{inc}}'(z(w))\right\} \frac{\partial z(w)}{\partial I}, \tag{58}$$

where $\frac{\partial x(w)}{\partial I}$ and $\frac{\partial z(w)}{\partial I}$ are the effects of additional disposable income on consumption of x and income; and λ is the multiplier on the government's budget constraint. These augmented weights incorporate both the direct social value of increasing the consumption of an individual of type w, and indirect impacts on social welfare from income effects.

The resulting optimal tax conditions for the two schedules are stated in Proposition 4. We defer the derivation of these conditions to Appendix C.

Proposition 4. *The optimal tax on the externality-producing good must satisfy Equation 59:*

$$\frac{\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + \mathcal{T}_x'\left(\hat{x}\left(z\right)\right)} = \eta_{x,z}^{\mathit{Taste}}\left(z\right)\frac{1}{\varepsilon_{x|z}\left(z\right)}\frac{1 - H_z\left(z\right)}{zh_z\left(z\right)}\left(1 - \bar{g}_+\left(z\right)\right), \tag{59}$$

and the optimal tax on income must satisfy Equation 60:

$$\frac{\mathcal{T}_z'\left(z\right) + \left[\mathcal{T}_x'\left(x\left(\hat{w}\left(z\right)\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x_{\mathit{inc}}'(z)}{1 - \mathcal{T}_z'\left(z\right)} = \frac{1}{\varepsilon_z\left(z\right)}\frac{1 - H_z\left(z\right)}{zh_z\left(z\right)}\left(1 - \bar{g}_+\left(z\right)\right), \tag{60}$$

where $\bar{g}_+\left(z\right)$ is the average marginal social welfare weight of agents with income above z.

Equation 59 closely resembles the standard "ABC" expression for the optimal nonlinear income tax for a redistributive social planner (Mirrlees, 1971). Part of the intuition is therefore familiar. A higher marginal tax rate at $\hat{x}\left(z\right)$ is optimal if: redistribution to agents with incomes below z is highly valued (i.e. if the average welfare weight above z, $\bar{g}_+\left(z\right)$, is relatively small); people do not respond much to the carbon tax (i.e. $\varepsilon_{x|z}\left(z\right)$ is small); and/or a large amount of revenue can be raised relative to the amount of income distorted by a local increase in the marginal tax rate ($1 - H_z(z)$ large relative to $zh_z(z)$).

There are two key differences from the standard optimal tax condition. The first is that there is a Pigouvian correction, which additively adjusts the marginal tax to take into account the marginal damage on others of consuming x, $\mathcal{D}'(\bar{x})$. The second is the presence of the taste elasticity, $\eta_{x,z}^{\text{Taste}}(z)$, which captures the same intuition as above.

Equation 60 is almost identical to the usual "ABC" formula. A higher marginal tax rate on income is optimal if redistribution is valued ($\bar{g}_{+}(z)$ is small), people do not respond much to the income tax ($\varepsilon_z(z)$ small), or a large amount of revenue can be raised relative to the amount of income distorted by a local increase in the marginal tax rate ($1 - H_z(z)$ large relative to $zh_z(z)$). The only difference from the standard intuition is the presence of $[\mathcal{T}_x'(x(\hat{w}(z))) - \mathcal{D}'(\bar{x})/\lambda]\, x_{\text{inc}}'(z)$ on the left hand side. As before, this term captures the intuition that marginal expenditure on the taxed good, x is socially beneficial if the carbon tax is set higher than the Pigouvian level; or socially costly otherwise.

C Proofs of Theoretical Results

Proof of Proposition 1. Consider the effects of an infinitesimal reform to the baseline tax schedules $\mathcal{T}_z(\cdot)$ and $\mathcal{T}_x(\cdot)$, as described in Section 2.4.

Behavioral Responses

Section 2.6 introduces notation for the various behavioral responses, but we define them more formally here. First, the compensated net-of-tax rate elasticity of taxable income is:

$$\varepsilon_z(w) \equiv -\frac{1 - \mathcal{T}_z'(z(w))}{\tau_z'(z(w))\, z(w)} \left. \frac{\partial z(w)}{\partial \kappa} \right|_{\kappa = \tau_z(z(w)) + \tau_x(x(w)) = \tau_x'(x(w)) = 0} \tag{61}$$

In words, this is the response of taxable income to a change to the marginal income tax rate without any change in the overall tax burden or the marginal tax rate on the externality-generating good.

Second, holding z constant, the compensated net-of-tax rate elasticity of spending on the externality-generating good is:

$$\varepsilon_{x|z}(w) \equiv -\frac{1 + \mathcal{T}_x'(x(w))}{\tau_x'(x(w))\, x(w)} \left. \frac{\partial x(w; z(w))}{\partial \kappa} \right|_{z(w), \kappa = \tau_z(z(w)) + \tau_x(x(w)) = \tau_z'(z(w)) = 0}. \tag{62}$$

Our assumptions ensure that all of these responses are smooth.

A Reform To Test Pareto Efficiency

To characterize the set of Pareto efficient tax systems, consider a distribution-neutral tax reform as presented in Section 2.5. The income tax is changed in a way that precisely offsets the impact of the change to the tax on the dirty good.

$$\tau_z(z) = -\tau_x(\hat{x}(z)). \tag{63}$$

This implies marginal income tax rate changes of

$$\tau_z'(z) = -\hat{x}'(z)\,\tau_x'(\hat{x}(z)), \tag{64}$$

Welfare Effects of a Tax Reform

This distribution-neutral reform has no mechanical effect on the utility level of any agent at any level of income. We argued in Section 2.9 that this leaves us with three effects that are local to the level of income at which agents face a change in their marginal tax rate, namely the following three components multiplied by $\tau_x'(\hat{x}(z))$:

$$\begin{aligned}
\Delta\mathcal{R}^{z}(z) &= \mathcal{T}_z'(z)\frac{z\varepsilon_z(z)}{1-\mathcal{T}_z'(z)}x_{\text{het}}'(z) \\
\Delta\mathcal{R}^{x|z}(z) &= -\left(\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)\frac{\varepsilon_{x|z}(z)\,\hat{x}(z)}{1+\mathcal{T}_x'(\hat{x}(z))} \\
\Delta\mathcal{R}^{z\to x}(z) &= \left(\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right)x_{\text{inc}}'(z)\frac{z\varepsilon_z(z)}{1-\mathcal{T}_z'(z)}x_{\text{het}}'(z)
\end{aligned}$$

where λ is the multiplier on the planner's budget constraint, and $\mathcal{D}'(\bar{x})/\lambda$ is the marginal externality cost associated with consumption of x.

Integrating over all types in the economy, the social welfare impact is:

$$\left.\frac{\partial\mathcal{L}}{\partial\kappa}\right|_{\kappa=0} = \int(\Delta\mathcal{R}^{z}(z) + \Delta\mathcal{R}^{x|z}(z) + \Delta\mathcal{R}^{z\to x}(z))\tau_x'(\hat{x}(z))\,h_z(z)\,\mathrm{d}z, \tag{65}$$

where $h_z(z)$ is the distribution of taxable income.

A Necessary Condition for Optimality

If the tax system is set optimally, it must be the case that any such infinitesimal tax reform has no first-order effect on welfare. That is to say, for any $\tau_x(\cdot)$ we must have $\left.\frac{\partial\mathcal{L}}{\partial\kappa}\right|_{\kappa=0} = 0$.

By the fundamental lemma of the calculus of variations,[22] we obtain equation 30, which we restate here for convenience

$$\frac{\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1+\mathcal{T}_x'(\hat{x}(z))} = \eta_{x,z}^{\text{Taste}}(z)\,\frac{\varepsilon_z(z)}{\varepsilon_{x|z}(z)}\left(\frac{\mathcal{T}_z'(z) + \left[\mathcal{T}_x'(\hat{x}(z)) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right]x_{\text{inc}}'(z)}{1-\mathcal{T}_z'(z)}\right)$$

[22]If for all continuous functions $\rho : \mathbb{R} \to \mathbb{R}$ we have $\int_0^\infty f(z)\rho(z)dz = 0$, and if f is continuous, then $f(z) = 0$ for all z. Letting $f(z) \equiv (\Delta\mathcal{R}^{z}(z) + \Delta\mathcal{R}^{x|z}(z) + \Delta\mathcal{R}^{z\to x}(z))h_z(z)$ and $\rho(z) \equiv \tau_x'(\hat{x}(z))$, equation 30 follows immediately.

It is straightforward to then re-arrange this to obtain 31, also restated:

$$\mathcal{T}_x'\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'\left(\bar{x}\right)}{\lambda} = \frac{\text{RR}(z)\mathcal{T}_z'\left(z\right)}{1-\text{RR}(z)x_{\text{inc}}'(z)},$$

where:

$$\text{RR}(z) \equiv \eta_{x,z}^{\text{Taste}}\left(z\right)\frac{\varepsilon_z\left(z\right)}{\varepsilon_{x|z}\left(z\right)}\frac{1+\mathcal{T}_x'\left(\hat{x}\left(z\right)\right)}{1-\mathcal{T}_z'\left(z\right)}.$$

Pareto Inefficiency

Because this reform holds each agent's total tax liability constant, it has no first-order impact on agents' private utility. Rather, if equation 26 is not equal to zero, the planner can raise revenue by moving in one direction or the other ($\kappa > 0$ or $\kappa < 0$) with no impact on individual utility. The resulting surplus in government revenue can then be used to provide a lump sum transfer, leading to a Pareto improvement. □

Proof of Proposition 4. Consider the effects of a more general infinitesimal reform to the baseline tax schedules $\mathcal{T}_z\left(\cdot\right)$ and $\mathcal{T}_x\left(\cdot\right)$, as described in Section 2.4. We start by introducing some additional notation, as well as characterizing the expanded set of behavioral responses to the reform, which our assumptions again ensure are smooth.

Behavioral Responses

In addition to the behavioral responses defined in our proof of Proposition 1, we will need notation for several additional responses. First, the income effects on taxable income and consumption of x are as follows:

$$\eta_z(w) \equiv -\frac{\left.\frac{\partial z(w)}{\partial \kappa}\right|_{\kappa=\tau_z'(z(w))=\tau_x'(x(w))=0}}{\tau_z\left(z\left(w\right)\right)+\tau_x\left(x\left(w\right)\right)} \quad \eta_{x|z}(w) \equiv -\frac{\left.\frac{\partial x(w;z(w))}{\partial \kappa}\right|_{z(w),\kappa=\tau_z'(z(w))=\tau_x'(x(w))=0}}{\tau_z\left(z\left(w\right)\right)+\tau_x\left(x\left(w\right)\right)}.$$

Using this notation, type w's overall responses to a reform are

$$\left.\frac{\mathrm{d}z\left(w\right)}{\mathrm{d}\kappa}\right|_{\kappa=0} = -\frac{\varepsilon_z\left(w\right)z\left(w\right)\tau_z'\left(z\left(w\right)\right)}{1-\mathcal{T}_z'\left(z\left(w\right)\right)} + \chi\left(z\left(w\right)\right)\tau_x'\left(x\left(w;z\right)\right) - \eta_z\left(w\right)\left[\tau_z\left(z\left(w\right)\right)+\tau_x\left(x\left(w;z\right)\right)\right] \tag{66}$$

$$\left.\frac{\mathrm{d}x\left(w;z\left(w\right)\right)}{\mathrm{d}\kappa}\right|_{\kappa=0} = -\frac{\varepsilon_{x|z}\left(w\right)x\left(w\right)}{1+\mathcal{T}_x'\left(x\left(w\right)\right)}\tau_x'\left(x\left(w\right)\right) - \eta_{x|z}\left(w\right)\left[\tau_z\left(z\left(w\right)\right)+\tau_x\left(x\left(w\right)\right)\right] + x_{\text{inc}}'(z(w))\left.\frac{\mathrm{d}z\left(w\right)}{\mathrm{d}\kappa}\right|_{\kappa=0} \tag{67}$$

where $\varepsilon_z(w)$ and $\varepsilon_{x|z}$ are as defined above, and:

$$\chi(w) \equiv \frac{1}{\tau_x'(x(w))} \left.\frac{\partial z(w)}{\partial \kappa}\right|_{\kappa=\tau_z(z(w))+\tau_x(x(w))=\tau_z'(z(w))=0}$$

is the cross-base response of taxable income to the commodity tax rate.

Inserting equation 66 into equation 67, we get:

$$\begin{aligned}\left.\frac{\mathrm{d}x(w;z(w))}{\mathrm{d}\kappa}\right|_{\kappa=0} =& \left[-\frac{\varepsilon_{x|z}(w)\,x(w)}{1+\mathcal{T}_x'(x(w))} + x_{\text{inc}}'(z(w))\chi(z(w))\right]\tau_x'(x(w)) \\ &- x_{\text{inc}}'(z(w))\frac{\varepsilon_z(w)\,z(w)}{1-\mathcal{T}_z'(z(w))}\tau_z'(z(w)) \\ &- \left[\eta_{x|z}(w) + x_{\text{inc}}'(z(w))\eta_z(w)\right]\left[\tau_z(z(w)) + \tau_x(x(w))\right].\end{aligned}$$

By Slutsky symmetry, we have the following expression for the partial effect of taxable income on the choice of the externality-generating good:

$$x_{\text{inc}}'(z(w)) = \frac{x(w)}{z(w)}\left[\frac{-\frac{1-\mathcal{T}_z'(z(w))}{x(w)}\chi(w)}{\varepsilon_z(w)}\right] = -\frac{1-\mathcal{T}_z'(z(w))}{z(w)}\frac{\chi(w)}{\varepsilon_z(w)}.$$

Put another way, we can write the cross-base response in terms of this partial effect:

$$\chi(w) = -\frac{z(w)\,\varepsilon_z(w)}{1-\mathcal{T}_z'(z(w))}x_{\text{inc}}'(z(w)).$$

Thus, the income response (equation 66) can be rewritten as

$$\begin{aligned}\left.\frac{\mathrm{d}z(w)}{\mathrm{d}\kappa}\right|_{\kappa=0} =& -\frac{z(w)\,\varepsilon_z(w)}{1-\mathcal{T}_z'(z(w))}\tau_z'(z(w)) - \frac{z(w)\,\varepsilon_z(w)}{1-\mathcal{T}_z'(z(w))}x_{\text{inc}}'(z(w))\tau_x'(x(w)) \\ &- \eta_z(w)\left[\tau_z(z(w)) + \tau_x(x(w))\right],\end{aligned} \tag{68}$$

and the response of x (equation 67) can be written as:

$$\begin{aligned}\left.\frac{\mathrm{d}x(w;z(w))}{\mathrm{d}\kappa}\right|_{\kappa=0} =& -\left[\frac{\varepsilon_{x|z}(w)\,x(w;z(w))}{1+\mathcal{T}_x'(x(w))} + \left(x_{\text{inc}}'(z(w))\right)^2\frac{z(w)\varepsilon_z(w)}{1-\mathcal{T}_z'(z(w))}\right]\tau_x'(x(w)) \\ &- x_{\text{inc}}'(z(w))\frac{z(w)\,\varepsilon_z(w)}{1-\mathcal{T}_z'(z(w))}\tau_z'(z(w)) \\ &- \left[\eta_{x|z}(w) + x_{\text{inc}}'(z(w))\eta_z(w)\right]\left[\tau_z(z(w)) + \tau_x(x(w))\right].\end{aligned} \tag{69}$$

Welfare Effects of a Tax Reform

For any given pair of continuously differentiable reform functions τ_z and τ_x, an infinitesimal reform away from $\mathcal{T}_z$ and $\mathcal{T}_x$ has the following effect on social welfare:

$$
\begin{aligned}
&\frac{1}{\lambda}\left.\frac{\partial\mathcal{W}}{\partial\kappa}\right|_{\kappa=0}+\left.\frac{\partial\mathcal{R}}{\partial\kappa}\right|_{\kappa=0} && (70)\\
&=\int\left(1-g\left(w\right)\right)\left[\tau_z\left(z\left(w\right)\right)+\tau_x\left(x\left(w\right)\right)\right]f\left(w\right)\mathrm{d}w && \text{(A)}\\
&-\int\left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right)+\left[\mathcal{T}_x'\left(x\left(w\right)\right)-\frac{\mathcal{D}'(\bar{x})}{\lambda}\right]x_{\text{inc}}'(z(w))}{1-\mathcal{T}_z'\left(z\left(w\right)\right)}\right)\varepsilon_z\left(w\right)z\left(w\right)\tau_z'\left(z\left(w\right)\right)f\left(w\right)\mathrm{d}w && \text{(B)}\\
&-\int\left\{\left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right)+\left[\mathcal{T}_x'\left(x\left(w\right)\right)-\frac{\mathcal{D}'(\bar{x})}{\lambda}\right]x_{\text{inc}}'(z(w))}{1-\mathcal{T}_z'\left(z\left(w\right)\right)}\right)x_{\text{inc}}'(z(w))\varepsilon_z\left(w\right)z\left(w\right)\ldots\right.\\
&\qquad\left.\ldots+\frac{\mathcal{T}_x'\left(x\left(w\right)\right)-\frac{\mathcal{D}'(\bar{x})}{\lambda}}{1+\mathcal{T}_x'\left(x\left(w\right)\right)}\varepsilon_{x|z}\left(w\right)x\left(w\right)\right\}\tau_x'\left(x\left(w\right)\right)f\left(w\right)\mathrm{d}w && \text{(C)}
\end{aligned}
$$

where λ is the multiplier on the planner's budget constraint, and $\mathcal{D}'\left(\bar{x}\right)/\lambda$ is the marginal externality cost associated with consumption of x.

The first line (A) captures the social welfare benefit of any redistribution from the reform. This includes differences in income effects and externality impacts, by the definition of $g(w)$. The second line (B) captures the fiscal externality via the income tax. The third line (C) captures the fiscal externality via the tax on x.

Optimality Condition for T_z

If the tax system is set optimally, it must be the case that any such infinitesimal tax reform has no first-order effect on welfare. That is to say, for any $\left(\tau_z\left(\cdot\right),\tau_x\left(\cdot\right)\right)$ we must have $\frac{1}{\lambda}\left.\partial\mathcal{W}/\partial\kappa\right|_{\kappa=0}+\left.\partial\mathcal{R}/\partial\kappa\right|_{\kappa=0}=0$. To characterize the optimal income tax, we can therefore consider a reform only to the income tax by supposing that $\tau_x\left(x\right)=0$ for all x, and then set equation 70 equal to zero.

With a change in variables, we can write the resulting optimality condition in terms of the income distribution. For all $\tau_z(\cdot)$:

$$
\begin{aligned}
&\int\left(1-g\left(\hat{w}\left(z\right)\right)\right)\tau_z\left(z\right)h_z\left(z\right)\mathrm{d}z\\
&\quad-\int\left[\frac{\mathcal{T}_z'\left(z\right)+\left[\mathcal{T}_x'\left(x\left(\hat{w}\left(z\right)\right)\right)-\frac{\mathcal{D}'(\bar{x})}{\lambda}\right]x_{\text{inc}}'(z)}{1-\mathcal{T}_z'\left(z\right)}\right]\varepsilon_z\left(\hat{w}\left(z\right)\right)z\tau_z'\left(z\right)h_z\left(z\right)\mathrm{d}z=0.
\end{aligned}
$$

By the fundamental lemma of the calculus of variations as we did for Proposition 1, we can then also write the following condition characterizing optimal marginal rates on income in terms of the distribution of income along with various other sufficient statistics:[23]

$$\frac{\mathcal{T}_z'(z) + \left[\mathcal{T}_x'(x(\hat{w}(z))) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x_{\text{inc}}'(z)}{1 - \mathcal{T}_z'(z)} = \frac{\int_z^\infty (1 - g(\hat{w}(s))) h_z(s)\, \mathrm{d}s}{\varepsilon_z(\hat{w}(z))\, z h_z(z)}. \tag{72}$$

Defining the average welfare weight for agents with income greater than z as:

$$1 - \bar{g}_+(z) = \frac{\int_z^\infty (1 - g(\hat{w}(s))) h_z(s)\, \mathrm{d}s}{1 - H_z(z)}, \tag{73}$$

we obtain the optimality condition stated in Proposition 4.

Optimality Condition for T_x

Next, to characterize the optimal tax on the externality-generating good, we consider a reform only to the tax on x by supposing that $\tau_z(z) = 0$ for all z, and set equation 70 to zero. Letting $H_x(x)$ be the distribution of x, with density $h_x(x)$, we can write the resulting optimality condition in terms of the distribution of consumption of x. For all $\tau_x(\cdot)$:

$$\begin{aligned}\int (1 - g(\tilde{w}(x)))\, \tau_x(x)\, h_x(x)\, \mathrm{d}x - \int \Bigg[\frac{\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + \mathcal{T}_x'(x)} \varepsilon_{x|z}(\tilde{w}(x))\, x \ldots \\ + \left(\frac{\mathcal{T}_z'(\tilde{z}(x)) + \left[\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x_{\text{inc}}'(\tilde{z}(x))}{1 - \mathcal{T}_z'(\tilde{z}(x))} \right) x_{\text{inc}}'(\tilde{z}(x)) \varepsilon_z(\tilde{w}(x)) \tilde{z}(x) \Bigg] \tau_x'(x) h_x(x) \mathrm{d}x = 0,\end{aligned}$$

where $\tilde{w}(x)$ and $\tilde{z}(x)$ are the values of w and z which correspond to each value of x.

Similar to the case of the income tax, the fundamental lemma of the calculus of variations then lets us write:

$$\begin{aligned}&\frac{\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + \mathcal{T}_x'(x)} \varepsilon_{x|z}(\tilde{w}(x))\, x h_x(x) \\ &\quad + x_{\text{inc}}'(\tilde{z}(x)) \left(\frac{\mathcal{T}_z'(\tilde{z}(x)) + \left[\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x_{\text{inc}}'(\tilde{z}(x))}{1 - \mathcal{T}_z'(\tilde{z}(x))} \right) \varepsilon_z(\tilde{w}(x)) \tilde{z}(x) h_x(x)\end{aligned} \tag{74}$$

[23]Here we use a slightly different variant of fundamental lemma of the calculus of variations, which states that, for all continuously differentiable functions $h : \mathbb{R}_+ \to \mathbb{R}$ we have

$$\int_0^\infty f(x)h(x)dx + \int_0^\infty g(x)h'(x)dx = 0, \tag{71}$$

and if f and g are continuous, then $g'(x) = f(x) \quad \forall x \geq 0$, g is continuously differentiable, and $\lim_{x\to\infty} g(x) = \lim_{x\to 0} g(x) = 0$. Our application of it here follows Jacquet and Lehmann (2021).

$$= \int_{x}^{\infty} \left(1 - g\left(\tilde{w}\left(y\right)\right)\right) h_{x}\left(y\right) dy.$$

We can then write equation 74 in terms of the distribution of taxable income. To do so, we use the fact that:

$$\begin{aligned} h_z\left(z\right) &= \frac{dx\left(\hat{w}\left(z\right);z\right)}{dz} h_x\left(x\left(\hat{w}\left(z\right);z\right)\right) \\ &= \left[x'_{\text{inc}}(z) + x'_{\text{het}}(z)\right] h_x\left(x\left(\hat{w}\left(z\right);z\right)\right). \end{aligned}$$

Using this result, multiplying both sides of equation 74 by $x'_{\text{inc}}(z) + x'_{\text{het}}(z)$, and inserting the optimality condition for the income tax (equation 72) we can rewrite equation 74 as:

$$\begin{aligned} &\frac{\mathcal{T}'_x\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + \mathcal{T}'_x\left(\hat{x}\left(z\right)\right)} \varepsilon_{x|z}\left(\hat{w}\left(z\right)\right) \hat{x}\left(z\right) h_z\left(z\right) + x'_{\text{inc}}(z) \int_z^{\infty} \left(1 - g\left(\hat{w}\left(s\right)\right)\right) h_z\left(s\right) ds \\ &\qquad = \left(x'_{\text{inc}}(z) + x'_{\text{het}}(z)\right) \int_z^{\infty} \left(1 - g\left(\hat{w}\left(s\right)\right)\right) h_z\left(s\right) ds, \end{aligned}$$

which further simplifies to

$$\frac{\mathcal{T}'_x\left(\hat{x}\left(z\right)\right) - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + \mathcal{T}'_x\left(\hat{x}\left(z\right)\right)} = x'_{\text{het}}(z) \frac{\int_z^{\infty} \left(1 - g\left(\hat{w}\left(s\right)\right)\right) h_z\left(s\right) ds}{\varepsilon_{x|z}\left(\hat{w}\left(z\right)\right) \hat{x}\left(z\right) h_z\left(z\right)}. \tag{75}$$

Using the definitions of $\eta^{\text{Taste}}_{x,z}\left(z\right)$ and $\bar{g}_{+}\left(z\right)$, we obtain the optimality condition stated in the proposition. □

Proof of Proposition 2. The proof here proceeds in a very similar manner to fully nonlinear case. Each agent now maximizes her utility subject to her slightly modified budget constraint, producing indirect utility $v(w)$.

$$\begin{aligned} v(w) = \max_{z,x,c} & \; u(c, x, z \mid w) \\ \text{s.t.} & \quad c + x + t_x x = z - T_z(z) \end{aligned} \tag{76}$$

Here, t_x is a linear tax on x.

A welfarist social planner chooses t_x and a twice-differentiable $\mathcal{T}_z$ to maximize social welfare while raising enough revenue to cover an exogenous revenue requirement, $\bar{\mathcal{R}}$.

$$\max \boldsymbol{\mathcal{W}} = \int \gamma(w) v(w) dF - \mathcal{D}(\overline{x}) \tag{77}$$

$$\text{s.t.} \quad \mathcal{R} \equiv \int \left(T_z(z(w)) + t_x x\right) dF(w) = \bar{\mathcal{R}} \tag{78}$$
$$\text{and} \quad \overline{x} = \int x(w) dF$$

Mathematically, everything remains identical to the proof of Proposition 1 up until the expression for the general impact on welfare of a tax reform (equation 70).

Imposing that the initial tax on x is linear yields equation 79.

$$\frac{1}{\lambda} \left.\frac{\partial \mathcal{W}}{\partial \kappa}\right|_{\kappa=0} + \left.\frac{\partial \mathcal{R}}{\partial \kappa}\right|_{\kappa=0} \tag{79}$$
$$= \int \left(1 - g\left(w\right)\right)\left[\tau_z\left(z\left(w\right)\right) + \tau_x\left(x\left(w\right)\right)\right] f\left(w\right) \mathrm{d}w \tag{A}$$
$$- \int \left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right) + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x'_{\text{inc}}(z(w))}{1 - \mathcal{T}_z'\left(z\left(w\right)\right)}\right) \varepsilon_z\left(w\right) z\left(w\right) \tau_z'\left(z\left(w\right)\right) f\left(w\right) \mathrm{d}w \tag{B}$$
$$- \int \left\{ \left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right) + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x'_{\text{inc}}(z(w))}{1 - \mathcal{T}_z'\left(z\left(w\right)\right)}\right) x'_{\text{inc}}(z(w))\varepsilon_z\left(w\right) z\left(w\right) \dots \right.$$
$$\left. \dots + \frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + t_x}\varepsilon_{x|z}\left(w\right) x\left(w\right) \right\} \tau_x'\left(x\left(w\right)\right) f\left(w\right) \mathrm{d}w \tag{C}$$

From here, we proceed similarly to the proof of Proposition 1.

However, the reform we consider must now be linear as well. Specifically, consider a joint reform where: $\tau_x\left(x\left(\hat{w}\left(z\right);z\right)\right) = \rho \times \hat{x}(z)$ for some constant ρ; and $\tau_z\left(z\right) = -\rho \times \hat{x}(z)$. For such a reform, we have $\tau_x'\left(x\left(\hat{w}\left(z\right);z\right)\right) = \rho$ and:

$$\tau_z'\left(z\right) = -\rho\left[x'_{\text{inc}}(z) + x'_{\text{het}}(z)\right]. \tag{80}$$

Combining that with equation 79, the impact on welfare is proportional to:

$$\int \left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right) + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x'_{\text{inc}}(z(w))}{1 - \mathcal{T}_z'\left(z\left(w\right)\right)}\right) \varepsilon_z\left(w\right) z\left(w\right) \left[x'_{\text{inc}}(z(w)) + x'_{\text{het}}(z(w))\right] f\left(w\right) \mathrm{d}w$$
$$- \int \left\{ \left(\frac{\mathcal{T}_z'\left(z\left(w\right)\right) + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x'_{\text{inc}}(z(w))}{1 - \mathcal{T}_z'\left(z\left(w\right)\right)}\right) x'_{\text{inc}}(z(w))\varepsilon_z\left(w\right) z\left(w\right) \dots \right.$$
$$\left. \dots + \frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + t_x}\varepsilon_{x|z}\left(w\right) x\left(w\right) \right\} f\left(w\right) \mathrm{d}w. \tag{81}$$

In this case, this simplifies to:

$$\int \left(\frac{\mathcal{T}_z' \left(z\left(w\right)\right) + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] x_{\text{inc}}'(z(w))}{1 - \mathcal{T}_z' \left(z\left(w\right)\right)} \right) \varepsilon_z \left(w\right) z\left(w\right) x_{\text{het}}'(z(w)) \cdots$$

$$\cdots - \left\{ \frac{t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}}{1 + t_x} \varepsilon_{x|z}\left(w\right) x\left(w\right) \right\} f\left(w\right) \mathrm{d}w. \tag{82}$$

To obtain the equation in the proposition, simply set this to zero to yield an optimality condition. Then replace the integrals with expectations over the income distribution, and impose the definition of $\mathrm{RR}(z)$.

By the same logic as before, there is no mechanical redistribution from this reform. It only impacts social welfare by changing government revenue or the externality. Thus, if this reform has a positive impact on social welfare, it is a Pareto improvement. □

Proof of Proposition 3. Setup and Behavioral Responses

Consider agents characterized by type (w, θ) where $w \in W \subseteq \mathbb{R}_{++}$ captures productivity and $\theta \in \Theta$ captures other preference heterogeneity. Each agent solves:

$$\begin{aligned} \max_{z,x,c} \quad & u(c, x, z; w, \theta) \\ \text{s.t.} \quad & c + x + \mathcal{T}_x(x) = z - \mathcal{T}_z(z) \end{aligned}$$

Define indirect utility $v(w, \theta)$ and choices $z(w, \theta)$, $x(w, \theta)$. The first-order conditions yield:

$$\begin{aligned} 1 + \mathcal{T}_x'(x) &= \mathrm{MRS}_x(c, x, z; w, \theta), \\ 1 - \mathcal{T}_z'(z) &= \mathrm{MRS}_z(c, x, z; w, \theta). \end{aligned}$$

For a reform parameterized by κ with $T_z(z;\kappa) = \mathcal{T}_z(z) + \kappa\tau_z(z)$ and $T_x(x;\kappa) = \mathcal{T}_x(x) + \kappa\tau_x(x)$, implicit differentiation of the first-order conditions yields the total behavioral responses. Decomposing into substitution and income effects:

$$\begin{aligned} \left.\frac{dz(w,\theta)}{d\kappa}\right|_{\kappa=0} = &-\frac{\varepsilon_z(w,\theta)z(w,\theta)\tau_z'(z(w,\theta))}{1 - \mathcal{T}_z'(z(w,\theta))} + \chi(w,\theta)\tau_x'(x(w,\theta)) \\ &- \eta_z(w,\theta)[\tau_z(z(w,\theta)) + \tau_x(x(w,\theta))], \end{aligned} \tag{83}$$

where $\varepsilon_z(w, \theta)$ is the compensated elasticity of taxable income, $\chi(w, \theta)$ is the cross-base response, and $\eta_z(w, \theta)$ is the income effect.

Slutsky Symmetry

By the symmetry of the Slutsky matrix, the cross-price effect of the commodity tax on labor supply equals the cross-price effect of the wage on commodity demand. This yields:

$$\chi(w,\theta) = -\frac{z(w,\theta)\varepsilon_z(w,\theta)}{1-\mathcal{T}_z'(z(w,\theta))}\frac{\partial x(w,\theta;z(w,\theta))}{\partial z}. \tag{84}$$

Substituting (84) into (83) and similarly deriving $dx/d\kappa$:

$$\begin{aligned}\left.\frac{dx(w,\theta)}{d\kappa}\right|_{\kappa=0} &= -\frac{\varepsilon_{x|z}(w,\theta)x(w,\theta)}{1+\mathcal{T}_x'(x(w,\theta))}\tau_x'(x(w,\theta)) \\ &\quad - \eta_{x|z}(w,\theta)[\tau_z(z)+\tau_x(x)] + \frac{\partial x}{\partial z}\left.\frac{dz}{d\kappa}\right|_{\kappa=0}.\end{aligned}$$

Welfare Effects of a Reform

Social welfare is $\mathcal{W} = \int_\Theta \int_W \gamma(w,\theta)v(w,\theta)f_w(w|\theta)dw\,\mu(d\theta) - \mathcal{D}(\bar{x})$. By the envelope theorem:

$$\left.\frac{dv(w,\theta)}{d\kappa}\right|_{\kappa=0} = -\frac{\partial v}{\partial c}[\tau_z(z(w,\theta))+\tau_x(x(w,\theta))].$$

Total revenue is $\mathcal{R} = \int_\Theta \int_W [T_z(z;\kappa)+T_x(x;\kappa)]f_w(w|\theta)dw\,\mu(d\theta)$. The Lagrangian $\mathcal{L} = \mathcal{W}/\lambda + \mathcal{R}$ responds to the reform as:

$$\begin{aligned}\left.\frac{d\mathcal{L}}{d\kappa}\right|_{\kappa=0} = &\underbrace{\int_\Theta\int_W (1-g(w,\theta))[\tau_z(z)+\tau_x(x)]f_w dw\,\mu(d\theta)}_{\text{Mechanical effect net of welfare loss}} \\ &+ \underbrace{\int_\Theta\int_W \mathcal{T}_z'(z)\left.\frac{dz}{d\kappa}\right|_{\kappa=0} f_w dw\,\mu(d\theta)}_{\text{Behavioral effect on income tax revenue}} \\ &+ \underbrace{\int_\Theta\int_W \left[\mathcal{T}_x'(x)-\frac{\mathcal{D}'(\bar{x})}{\lambda}\right]\left.\frac{dx}{d\kappa}\right|_{\kappa=0} f_w dw\,\mu(d\theta)}_{\text{Behavioral effect on commodity tax revenue}},\end{aligned}$$

where the social marginal utility of income is:

$$g(w,\theta) \equiv \frac{\gamma(w,\theta)}{\lambda}\frac{\partial v}{\partial c} + \mathcal{T}_z'(z)\eta_z(w,\theta) + \left[\mathcal{T}_x'(x)-\frac{\mathcal{D}'(\bar{x})}{\lambda}\right]\left[\eta_{x|z}(w,\theta)+\frac{\partial x}{\partial z}\eta_z(w,\theta)\right].$$

Change of Variables to (z,x)

Assume that, conditional on θ, $z(w,\theta)$ is continuous and monotonic in w. This ensures a continuous marginal distribution of z with density $h_z(z)$. However, the conditional distribution of x given z may have mass points, so we work with the CDF $H_x(x|z) = P(x(w,\theta) \leq x \mid z(w,\theta) = z)$ and corresponding measure $\nu_x(A|z) = \int_{x \in A} dH_x(x|z)$.

Define $\bar{\ell}(z,x) \equiv \mathbb{E}[\ell(w,\theta) \mid z(w,\theta) = z, x(w,\theta) = x]$ for $\ell \in \{g, \varepsilon_z, \varepsilon_{x|z}\}$. The welfare effect becomes:

$$
\begin{aligned}
\left.\frac{d\mathcal{L}}{d\kappa}\right|_{\kappa=0} = & \int_{\mathbb{R}_{++}} \int_{\mathbb{R}_{++}} (1 - \bar{g}(z,x))[\tau_z(z) + \tau_x(x)]\, dH_x(x|z)\, h_z(z)\, dz \\
& - \int_{\mathbb{R}_{++}} \int_{\mathbb{R}_{++}} \frac{\mathcal{T}_z'(z)\bar{\varepsilon}_z(z,x) + \left[\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] \mathbb{E}[x_{\text{inc}}' \varepsilon_z | z, x]}{1 - \mathcal{T}_z'(z)} z\tau_z'(z)\, dH_x(x|z)\, h_z(z)\, dz \\
& - \int_{\mathbb{R}_{++}} \int_{\mathbb{R}_{++}} \left\{ \frac{\mathcal{T}_z'(z)\mathbb{E}[x_{\text{inc}}' \varepsilon_z | z, x] + \left[\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] \mathbb{E}[(x_{\text{inc}}')^2 \varepsilon_z | z, x]}{1 - \mathcal{T}_z'(z)} z \right. \\
& \qquad \left. + \frac{\left[\mathcal{T}_x'(x) - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] \bar{\varepsilon}_{x|z}(z,x) x}{1 + \mathcal{T}_x'(x)} \right\} \tau_x'(x)\, dH_x(x|z)\, h_z(z)\, dz.
\end{aligned}
$$

Vertically Neutral Reform

Consider a linear tax reform $\tau_x(x) = x$ with compensating income tax changes $\tau_z(z) = -\mathbb{E}[\tau_x(X)|Z = z] = -\bar{x}(z)$, which holds average tax payments constant at each income level. This yields $\tau_z'(z) = -\bar{x}'(z)$.

Preference Heterogeneity Sufficient Statistic

Define the preference heterogeneity sufficient statistic:

$$
x_{het}'(z;\theta) \equiv \bar{x}'(z) - x_{\text{inc}}'(z;\theta).
$$

This is the residual of the cross-sectional relationship between x and z after removing the causal income effect for type θ agents.

Optimal Tax Condition

For a linear tax $\mathcal{T}_x(x) = t_x x$, substituting the vertically neutral reform into the welfare effect and setting $\left.\frac{d\mathcal{L}}{d\kappa}\right|_{\kappa=0} = 0$ yields:

$$
\begin{aligned}
0 = & - \int_{\mathbb{R}_{++}} \mathbb{C}[g(z;\theta), x(z;\theta)|z] h_z(z)\, dz \\
& + \int_{\mathbb{R}_{++}} \frac{\mathcal{T}_z'(z)}{1 - \mathcal{T}_z'(z)} \mathbb{E}[x_{het}'(z;\theta)\varepsilon_z(z;\theta)|z] z h_z(z)\, dz \\
& + \left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right] \int_{\mathbb{R}_{++}} \frac{1}{1 - \mathcal{T}_z'(z)} \mathbb{E}[x_{het}'(z;\theta) x_{inc}'(z;\theta)\varepsilon_z(z;\theta)|z] z h_z(z)\, dz
\end{aligned}
$$

$$- \frac{\left[t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda}\right]}{1+t_x} \mathbb{E}[x(z;\theta)\varepsilon_{x|z}(z;\theta)].$$

Under the assumption that $\mathbb{E}[\mathbb{C}(g(w,\theta), x(w,\theta)|z)] = 0$, the optimal tax satisfies equation (47):

$$t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda} = \frac{\mathbb{E}_{z,\theta}[\text{RR}(z;\theta)\mathcal{T}_z'(z)]}{1 - \mathbb{E}_{z,\theta}[\text{RR}(z;\theta)x'_{\text{inc}}(z;\theta)]},$$

where $\text{RR}(z;\theta) = \eta^{\text{Taste}}_{x,z}(z;\theta)\left(\frac{\varepsilon_z(z;\theta)\bar{x}(z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\right)\left(\frac{1+t_x}{1-\mathcal{T}_z'(z)}\right)$.

□

Proof of Corollary 1. Suppose $\varepsilon_z(z;\theta) = \bar{\varepsilon}_z(z)$ for all θ at each z. Then the covariance terms in the numerator and denominator of the general optimal tax condition simplify.

For the numerator, $\mathbb{C}[\varepsilon_z(z;\theta), x'_{inc}(z;\theta)|z] = 0$ by assumption, leaving only the average terms. For the denominator, using $x'_{het} = \bar{x}' - x'_{inc}$:

$$\begin{aligned}
\mathbb{E}[x'_{\text{het}}x'_{\text{inc}}\varepsilon_z|z] &= \bar{\varepsilon}_z(z)\mathbb{E}[(\bar{x}'(z) - x'_{\text{inc}}(z;\theta))x'_{\text{inc}}(z;\theta)|z] \\
&= \bar{\varepsilon}_z(z)\left[\bar{x}'(z)\bar{x}'_{\text{inc}}(z) - \mathbb{E}[(x'_{\text{inc}})^2|z]\right] \\
&= \bar{\varepsilon}_z(z)\left[\bar{x}'(z)\bar{x}'_{\text{inc}}(z) - (\bar{x}'_{\text{inc}}(z))^2 - \mathbb{V}[x'_{\text{inc}}|z]\right] \\
&= \bar{\varepsilon}_z(z)\bar{x}'_{\text{het}}(z)\bar{x}'_{\text{inc}}(z) - \bar{\varepsilon}_z(z)\mathbb{V}[x'_{\text{inc}}|z].
\end{aligned}$$

Substituting into the general formula and collecting terms yields equation (50):

$$t_x - \frac{\mathcal{D}'(\bar{x})}{\lambda} = \frac{\mathbb{E}_z[\overline{\text{RR}}(z)\mathcal{T}_z'(z)]}{1 - \mathbb{E}_z[\overline{\text{RR}}(z)\bar{x}'_{\text{inc}}(z)] + \mathbb{E}_z[\widetilde{\text{RR}}(z)\mathbb{V}(x'_{\text{inc}}(z;\theta)|z)]},$$

where:

$$\overline{\text{RR}}(z) = \bar{\eta}^{\text{Taste}}_{x,z}(z)\frac{\bar{\varepsilon}_z(z)\bar{x}(z)}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\frac{1+t_x}{1-\mathcal{T}_z'(z)},$$

$$\widetilde{\text{RR}}(z) = \frac{\bar{\varepsilon}_z(z)z}{\mathbb{E}[\varepsilon_{x|z}(z;\theta)x(z;\theta)]}\frac{1+t_x}{1-\mathcal{T}_z'(z)}.$$

The variance term $\mathbb{V}[x'_{\text{inc}}|z]$ appears in the denominator with a positive sign, which attenuates any deviation from Pigouvian taxation. Only when $\mathbb{V}[x'_{\text{inc}}|z] = 0$ for all z does the formula reduce to the unidimensional case. □

D Survey Questionnaire Appendix

D.1 Survey Link and Metadata

Survey link: https://umich.qualtrics.com/jfe/form/SV_egEUDcSVdWIjkTY

Survey flow: Landing Screen; Actual Expenditure Block; Attention Check; Causal Response Elicitation; Demographics; Commute Block.

The questionnaire below reproduces the respondent-facing text, answer options, and relevant display logic. Administrative Qualtrics timing fields and embedded Prolific/session fields are omitted; graphic-only display elements are noted where they appear.

D.2 Prolific-Provided Respondent Demographics

Respondents were recruited through Prolific, and the survey records were linked to Prolific profile data using the Prolific participant IDs.

- **Age:** integer age in years.
- **Sex:** Female; Male.
- **Ethnicity:** Asian; Black; Mixed; Other; White.
- **Country of birth, country of residence, and nationality:** country fields from the respondent's Prolific profile.
- **Language:** respondent language recorded by Prolific.
- **Student status:** Yes; No.
- **Employment status:** Full-Time; Part-Time; Unemployed (and job seeking); Not in paid work (e.g. homemaker, retired or disabled); Due to start a new job within the next month.

D.3 Survey Questions

Landing Screen

1. **Welcome**

   Thank you for your interest in taking this survey. Your participation is valuable for our research, which will help understand how people make spending decisions.

   Before proceeding, please review our Participant Information Sheet.

   By moving forward, you confirm that:

   1. You have read and understood the Participant Information Sheet
   2. You voluntarily agree to participate in this research
   3. You understand you can withdraw at any time

   Click → to begin the survey.

Actual Expenditure Block

1. Please indicate your total **annual** household income **before** taxes paid to the government, transfers received from the government, and deductions in the current fiscal year (2024). Please include income from wages, salaries, self-employment, investments, social security, retirement accounts, and any other sources.

   *If you are not sure, it is okay to provide your best estimate. However, an accurate estimate of your **annual** before-tax income would be very valuable for our research.*

   Response format: Open numeric entry, whole dollars, minimum 0. The live page displays implied household monthly before-tax income from the annual entry.

2. Please indicate your total **annual** household income **after** taxes paid to the government, transfers received from the government, and deductions in the current fiscal year (2024). You can think of this as your "take-home" income.

   *If you are not sure, it is okay to provide your best estimate. To reiterate, please enter your **annual** after-tax income, but note that your corresponding **monthly** household "take-home" income will automatically be displayed below.*

   Response format: Open numeric entry, whole dollars, minimum 0. The live page displays implied household monthly after-tax income from the annual entry.

   Display element: The survey displays a graphic-only expenses icon before the next item.

3. In the rest of the survey, we will ask you about your consumption across different expenditure categories such as food, housing expenditures, or utilities. But for now, we want to know whether you are paying attention, so please select "Healthcare (health insurance, medical services, drugs, and medical supplies)."

   Response options:

   – Food (both at and away from home) and beverages (including alcoholic beverages)
   – Water, sewage, and other public services (e.g. trash and garbage collection)
   – Gasoline, diesel and/or other motor fuels
   – Housing expenditure excluding utilities (includes rent, owned dwelling expenditures, mortgage payments, home insurance, furnishing and appliances, and domestic services)
   – Healthcare (health insurance, medical services, drugs, and medical supplies)
   – Electricity, natural gas (for heating and cooking), fuel oil and other home heating fuels

4. How much of your household's [monthly after-tax income amount] **monthly after-tax income** did you spend (as opposed to save) on each of the following expenditure categories, on average over the last 3 months?

   These categories, including savings, are exhaustive, and the total amount (at the bottom) will automatically sum up to your household's total monthly after-tax income. Your household's average monthly savings will automatically be calculated and displayed below.

Response format: Constant-sum dollar entries by category. Savings and total are displayed and calculated on screen. The live page displays household average monthly savings and uses a "Validate and Continue" button.

Note: The QSF/PDF placeholder is "MonthExp"; the live survey replaces it with the monthly dollar amount implied by the respondent's annual after-tax income.

- Food (both at and away from home) and beverages (including alcoholic beverages): _____
- Housing expenditure excluding utilities (includes rent, owned dwelling expenditures, mortgage payments, home insurance, furnishing and appliances, and domestic services): _____
- Electricity, natural gas (for heating and cooking), fuel oil and other home heating fuels: _____
- Water, sewage, and other public services (e.g. trash and garbage collection): _____
- Healthcare (health insurance, medical services, drugs, and medical supplies): _____
- Gasoline, diesel and/or other motor fuels: _____
- Flights / Airfare expenditures: _____
- All other categories (telecommunications services, entertainment, other types of transportation, clothing, personal insurance and pensions, and education expenditures): _____
- Savings: _____
- Total: _____

Attention Check

Display element: The survey displays the permanent-income-increase graphic before the questions in this block. The same graphic appears again before the causal response elicitation question; Figure D.1 shows that screen.

1. Earlier you told us your after-tax household income last year was [annual after-tax income].

   Now imagine that you or someone in your household receives an **unexpected, permanent pay raise** (or a similar windfall such as a new government benefit, or a tax credit) that **boosts your after-tax income by exactly $1,000** forever.

   - Starting now your after-tax income becomes [annual after-tax income] **+ $1,000**
   - Every future year it will be **$1,000 higher than whatever amount you would otherwise have expected**.

   In this scenario, will your household income be higher **three years from now** than what you originally expected?

   Note: The QSF/PDF placeholder is "AnnAfterTaxInc"; the live survey replaces it with the respondent's annual after-tax income.

   Response options:

   - Yes
   - No

2. By how much has your household income increased in this scenario, compared to what you originally expected?

   Response options:

   – $500 of total lifetime income
   – $1,000 of total lifetime income
   – $500 each year
   – $1,000 each year

Causal Response Elicitation

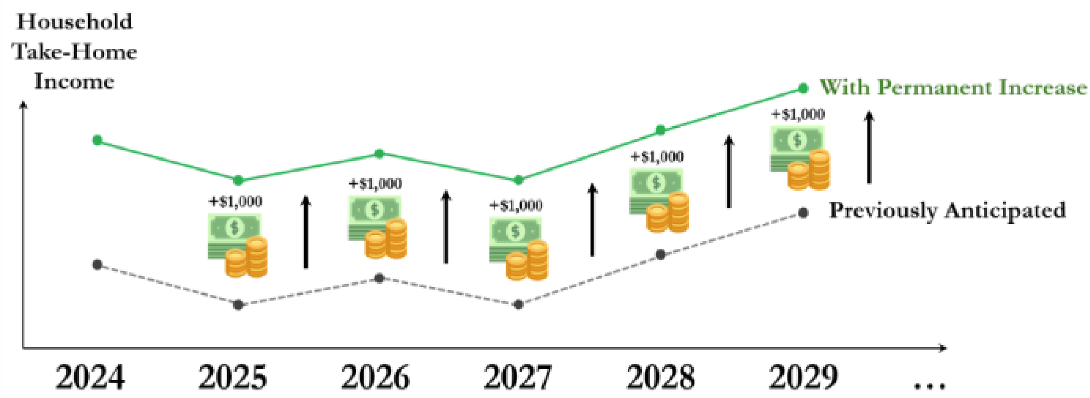
Display element: The survey displays the permanent-income-increase graphic immediately above the following question.

1. Imagine once again that your after-tax household's income is $1,000 higher **each year** than what you originally expected, forever. On **average**, how much of this extra $1,000 would your household spend (rather than save) on the following categories, **each year**?

   For each category, indicate a number corresponding to the additional dollar amount spent on each category. These categories, including savings, are exhaustive, and the total amount (at the bottom) will automatically sum up to $1,000. The amount saved each year will automatically be calculated and displayed below.

   Response format: Constant-sum dollar entries by category. Savings and total are displayed and calculated on screen, with the total summing to $1,000. The live page displays the additional amount saved each year and uses a "Validate and Continue" button.

   – Food (both at and away from home) and beverages (including alcoholic beverages): _____
   – Housing expenditure excluding utilities (includes rent, owned dwelling expenditures, mortgage payments, home insurance, furnishing and appliances, and domestic services): _____
   – Electricity, natural gas (for heating and cooking), fuel oil and other home heating fuels: _____
   – Water, sewage, and other public services (e.g. trash and garbage collection): _____
   – Healthcare (health insurance, medical services, drugs, and medical supplies): _____
   – Gasoline, diesel and/or other motor fuels: _____
   – Flights / Airfare expenditures: _____
   – All other categories (telecommunications services, entertainment, other types of transportation, clothing, personal insurance and pensions, and education expenditures): _____
   – Savings: _____
   – Total: _____



Imagine once again that your after-tax household's income is $1,000 higher **each year** than what you originally expected, forever. On **average**, how much of this extra $1,000 would your household spend (rather than save) on the following categories, **each year**?

For each category, indicate a number corresponding to the additional dollar amount spent on each category. These categories, including savings, are exhaustive, and the total amount (at the bottom) will automatically sum up to $1,000. The amount saved each year will automatically be calculated and displayed below.

Food (both at and away from home) and beverages (including alcoholic beverages)	0
Housing expenditure excluding utilities (includes rent, owned dwelling expenditures, mortgage payments, home insurance, furnishing and appliances, and domestic services)	0
Electricity, natural gas (for heating and cooking), fuel oil and other home heating fuels	0
Water, sewage, and other public services (e.g. trash and garbage collection)	0
Healthcare (health insurance, medical services, drugs, and medical supplies)	0
Gasoline, diesel and/or other motor fuels	0
Flights / Airfare expenditures	0
All other categories (telecommunications services, entertainment, other types of transportation, clothing, personal insurance and pensions, and education expenditures)	0
Savings	1000
Total	1,000

Additional amount **saved each year**: $1,000

Figure D.1. Permanent-income-increase graphic and survey screen for the causal consumption response elicitation block

Demographics

1. How would you describe the area where you live?

   Response options:

   – Urban (e.g., city or a large town)
   – Suburban (e.g., a residential area near a city)
   – Rural (e.g., a countryside, a small town, or a village)

2. In which state do you currently reside?

   Response format: Dropdown menu: Alabama; Alaska; Arizona; Arkansas; California; Colorado; Connecticut; Delaware; District of Columbia; Florida; Georgia; Hawaii; Idaho; Illinois; Indiana; Iowa; Kansas; Kentucky; Louisiana; Maine; Maryland; Massachusetts; Michigan; Minnesota; Mississippi; Missouri; Montana; Nebraska; Nevada; New Hampshire; New Jersey; New Mexico; New York; North Carolina; North Dakota; Ohio; Oklahoma; Oregon; Pennsylvania; Rhode Island; South Carolina; South Dakota; Tennessee; Texas; Utah; Vermont; Virginia; Washington; West Virginia; Wisconsin; Wyoming; Outside the US

3. What is the highest level of education you have completed or the highest degree you have received?

   Response options:

   – High school, high school equivalent, or lower
   – Some college, no degree
   – Associate's degree in college (e.g., AA, AS)
   – Bachelor's degree (e.g., BA, BS)
   – Masters (MA, MS, MEd), professional (JD, MD, DDS) or doctorate degree (PhD, EdD)

4. Please indicate your agreement with the following statement: I spend my entire income on paper clips.

   Response options:

   – Strongly Disagree
   – Disagree
   – Agree
   – Strongly Agree

5. Which category best represents your household?

   Response options:

   – Married Couple only
   – Married Couple, with children
   – One parent, with children
   – Single, without children
   – Other (please specify) ________

6. How many children are part of your household and younger than 18 years of age?

   Response format: Open numeric entry, integer from 0 to 50.

   Note: Displayed if the respondent selected Married Couple, with children; One parent, with children; or Other (please specify) in the household-type question.

7. How many of these children part of your household are 5 years or younger?

   Response format: Open numeric entry, integer from 0 to 50. The live page displays a reminder that the value should be no more than the reported number of children under 18 and enforces that cap.

   Note: Displayed if the respondent selected Married Couple, with children; One parent, with children; or Other (please specify) in the household-type question.

8. What is your spouse/partner's age? Enter a number only, in years.

   Response format: Open numeric entry, integer from 1 to 130.

   Note: Displayed if the respondent selected Married Couple only; Married Couple, with children; or Other (please specify) in the household-type question.

Commute Block

1. How far is your commute (one-way) from your home to your work location? Enter your best estimate of the **distance in miles**. If you do not work, or work from home, simply enter 0.

   Response format: Open numeric entry, minimum 0.

2. What is your primary mode of transportation/commuting?

   Response options:

   – Car
   – Public transportation (bus/train)
   – Ride-sharing app
   – Bicycle
   – Walking
   – Other (please specify) ________

3. How easy or difficult would it be for you to switch to a more environmentally friendly mode of transportation (one that produces less carbon emissions)?

   Response options:

   – Very Difficult
   – Difficult
   – Nor Easy, nor Difficult
   – Easy
   – Very Easy